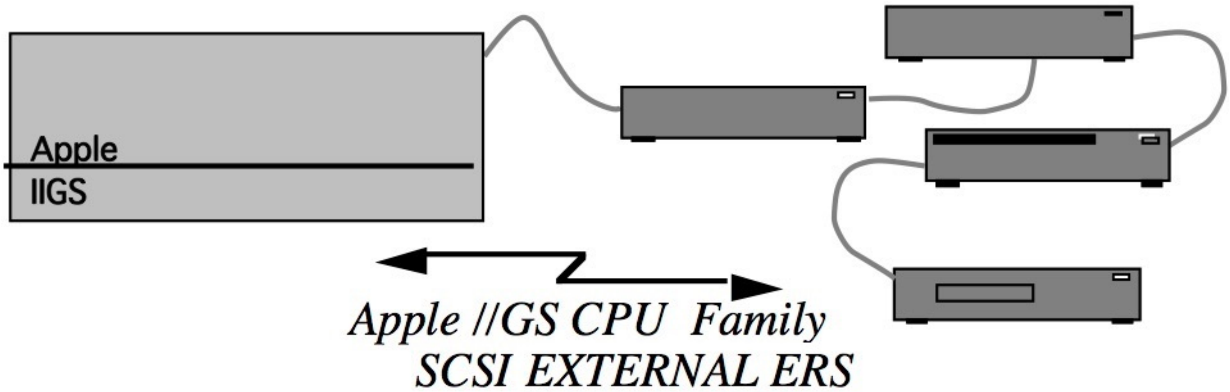
Apple SCSI Scanners
ERS
Version 1.0



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Technical specifications

Apple Scanner (A9M0337)

Scanner type	Flatbed
Maximum document size	8.5-inch by 14.0-inch
Interface	SCSI
Composition types	Line Art, Halftone, Grayscale
Grayscale	16 levels (4 bits)
Output resolution	75 - 300 dpi
Contrast levels	8
Brightness levels	16
Threshold values	16
Graymap settings	3
Halftone patterns	Spiral, Bayer; downloaded patterns may be up to 4-by-4 matrix

Apple OneScanner (M1381)

Scanner type	Flatbed
Maximum document size	8.5-inch by 14.0-inch
Interface	SCSI
Composition types	Line Art, Halftone, Grayscale
Grayscale	256 levels (8 bits)
Output resolution	72 - 300 dpi in 1 dpi increments
Contrast levels	256
Brightness levels	256
Threshold values	256
Graymap settings	256
Halftone patterns	4-by-4 and 8-by-8 Spiral and Bayer patterns built in; downloaded patterns may range from 1-by-2 to 16-by-16

Apple Color Scanner (BO993LL/A)

Scanner type	Flatbed
Maximum document size	8.5-inch by 14.0-inch
Interface	SCSI
Composition types	Line Art, Grayscale, Bi-level Color mode, Full Color mode
Color	16 million colors (24 bits)
Output resolution	75 - 1200 dpi
Contrast levels	256
Brightness levels	256
Threshold values	256
Graymap settings	256
Halftone patterns	Not supported

Apple Color OneScanner 600/27 (M4496LL/A)

Scanner type	Flatbed
Maximum document size	8.5-inch by 11.5-inch
Interface	SCSI
Composition types	Line Art, Grayscale, Bi-level Color mode, Full Color mode
Color	16 million colors externally (134 million colors externally, 27 bits)
Output resolution	300 by 600 dots per inch
Contrast levels	256
Brightness levels	256
Threshold values	256
Graymap settings	256
Halftone patterns	Not supported

Apple Color OneScanner 1200/30 (M4495LL/A)

Scanner type	Flatbed
Maximum document size	8.5-inch by 11.5-inch
Interface	SCSI
Composition types	Line Art, Grayscale, Bi-level Color mode, Full Color mode
Color	1 billion colors (30 bits)
Output resolution	600 by 1200 dots per inch
Contrast levels	256
Brightness levels	256
Threshold values	256
Graymap settings	256
Halftone patterns	Not supported

Enable Vital Product data
Some of the scanners contain a vital product page. To get the data, send an INQUIRY command with the EVPD bit set to 1, and the page code equals to $F0. The resulting DStatus data is 12 01 F0 00 F0 00. The length of returned data depends on the scanner:

- Scanner
  Not applicable

- OneScanner
  Not applicable

- Color OneScanner
  Not applicable

- Color OneScanner 600/27, length is $45 and data is

```
06 F0 02 00 45 01 2C 01 2C 11 02 58 02 58 00 0C
00 0C FF FC 00 00 09 F8 00 00 0D A8 0E 00 00 00
09 F8 00 00 10 68 40 01 01 0D 0A FF DF 60 00 00
03 00 08 C1 88 10 00 00 01 00 08 00 0E 04 01 00
00 00 00 8D 00 00 80 00 00 00 FF 00 00 00 00 00
```

- Color OneScanner 1200/30, length is $45 and data is

```
06 F0 02 00 45 01 2C 01 2C 11 04 B0 04 B0 00 0C
00 0C FF FF 00 00 13 F0 00 00 1B 50 0E 00 00 00
13 F0 00 00 20 D0 40 01 01 0D 0A FF DF 60 00 00
03 00 08 C1 88 10 00 00 01 00 08 00 0E 04 01 00
00 00 00 8D 00 02 00 00 00 00 FF 00 00 00 00 00
```

Notes
The scanners use a default measurement unit divisor of 1200, ie. the 1200ths of an inch.
All values are given in decimal.

SCSI commands

TEST UNIT READY ($00)

The TEST UNIT READY command provides a means for the host computer to check whether the scanner is ready for normal operation. If the scanner is in a ready condition, the command reports a GOOD status. If the scanner is not ready, the command sets the SCSI sense data to the cause of the error, and the command is terminated with a CHECK CONDITION status. To retrieve the sense data, issue the REQUEST SENSE command (described later in this chapter). Figure 7-1 shows the format of the TEST UNIT READY command structure.

This command is not a request for self-testing. For more information about internal scanner tests, see "SEND DIAGNOSTIC ($1D)," later in this chapter.

The TEST UNIT READY command provides a means to check if the logical unit is ready. This is not a request for a self-test. If the logical unit would accept an appropriate medium-access command without returning CHECK CONDITION status, this command shall return a GOOD status. If the logical unit cannot become operational or is in a state such that an initiator action (e.g. START UNIT command) is required to make the unit ready, the target shall return CHECK CONDITION status with a sense key of NOT READY.

Bit Byte	7	6	5	4	3	2	1	0
0	($00) Operation code							
1	Logical unit number			($00) Reserved				
2	($00) Reserved							
3	($00) Reserved							
4	Allocation length							
5	Control							

Table 74 defines the preferred responses to the TEST UNIT READY command. Higher-priority responses (e.g. BUSY or RESERVATION CONFLICT) are also permitted.

Table 74 - Preferred TEST UNIT READY responses

Status	Sense key	ASC and ASCQ
GOOD	NO SENSE	NO ADDITIONAL SENSE INFORMATION or other valid additional sense code.
CHECK CONDITION	ILLEGAL REQUEST	LOGICAL UNIT NOT SUPPORTED
CHECK CONDITION	NOT READY	LOGICAL UNIT DOES NOT RESPOND TO SELECTION
CHECK CONDITION	NOT READY	MEDIUM NOT PRESENT
CHECK CONDITION	NOT READY	LOGICAL UNIT NOT READY, CAUSE NOT REPORTABLE
CHECK CONDITION	NOT READY	LOGICAL UNIT IS IN PROCESS OF BECOMING READY
CHECK CONDITION	NOT READY	LOGICAL UNIT NOT READY, INITIALIZATION COMMAND REQUIRED
CHECK CONDITION	NOT READY	LOGICAL UNIT NOT READY, MANUAL INTERVENTION REQUIRED
CHECK CONDITION	NOT READY	LOGICAL UNIT NOT READY, FORMAT IN PROGRESS

REQUEST SENSE ($03)

The REQUEST SENSE command instructs the scanner to return error and status information. The SCSI sense data is valid after any operation that returns a CHECK CONDITION status. If another operation returns a CHECK CONDITION status before your program retrieves the current sense data, the scanner overwrites the old sense data with the new data.

Figure 7-2 shows the format of the REQUEST SENSE command. The returned information is sent to the host computer in a return structure appropriate to the type of attached scanner. The return structures for the Apple Scanner, OneScanner, and Color OneScanner are described later in this section.

Bit Byte	7	6	5	4	3	2	1	0
0	($03) Operation code							
1	Logical unit number			($00) Reserved				
2	($00) Reserved							
3	($00) Reserved							
4	Allocation length							
5	Control							

REQUEST SENSE Return Structure Description for the Apple Scanner

The REQUEST SENSE command returns a structure containing the requested information. Figure 7-3 shows the format of the return structure for the Apple Scanner.

Figure 7-3 The REQUEST SENSE return structure for the Apple Scanner

Bit / Byte	7	6	5	4	3	2	1	0
0	($00) Reserved	(7)			($00) Reserved			
1	($00) Reserved							
2	($00) Reserved				Sense key			
3 - 6	($00) Reserved							
7	Additional sense length							
8 - 17	($00) Reserved							
18	Dim light	No light	No home	No limit	Shade error	ROM error	RAM error	CPU error
19	DIPP error	DMA error	GA1 error	($00) Reserved				
20	Optical gain							
21	($00) Reserved							

Here is a list of the fields and bits defined in this return structure:

Reserved These fields are reserved for future expansion. Set them to 0.

Sense key Bits 0–3 of byte 2 contain the Sense key. The possible Sense key values are as follows:

- $0 No sense information: there is no specific sense data to be returned to the host computer. This situation occurs after a command executes successfully.
- $2 Not ready: the scanner cannot scan without some user intervention.
- $4 Hardware error: the scanner detects a nonrecoverable hardware fault while executing a command or during a self-test; these errors are indicated by the setting of the hardware error flags in bytes 18 and 19 of the return data block.
- $5 Illegal request: the scanner detects an illegal parameter in the command block; the scanner cannot perform a scan and cannot alter any parameter data within the scanner.
- $6 Unit attention: the scanner has just been switched on or reset and cannot execute any command except REQUEST SENSE or INQUIRY.
- $9 Vendor unique: the scanner detects a vendor unique error, which is identified by the setting of the Vendor unique error flags in byte 18 of the return data block.

Additional sense length

This field indicates the number of sense data bytes that follow. The value of this field does not include bytes 0–7 of the return structure. It reflects the maximum size possible for a returned structure. The scanner does not adjust the value of this field to accommodate the Transfer length parameter specified in your REQUEST SENSE command.

Hardware error flags

Bytes 18 and 19 in the return structure contain bit flags that indicate the specific nature of a hardware error. Flags that are set to 1 indicate a specific hardware error condition. If the returned Sense key field is set to $4, then one or more of these bits are set to 1.

No light	No light is detected from the fluorescent lamp.
No limit	The scanner was unable to locate the limit switch position. Scanning cannot proceed because physical damage to the device may occur.
ROM error	A fatal hardware error has been detected in the ROM (an erroneous checksum).
RAM error	A fatal hardware error has been detected in the RAM.
CPU error	A fatal hardware error has been detected in the CPU.
DIPP error	A fatal hardware error has been detected in the document image preprocessor (DIPP) IC.
DMA error	A fatal hardware error has been detected in the direct memory access (DMA) IC.
GA1 error	A fatal hardware error has been detected in the Gate Array 1 (GA1) IC.

Vendor unique error flags

Byte 18 in the return structure contains bit flags that indicate the specific nature of a vendor unique error. Flags that are set to 1 indicate a specific error condition. If the returned Sense key field is set to $9, then one or more of these bits is set to 1.

Dim light	The fluorescent lamp is functioning, but its output has dropped below 70 percent and is too low for a scan to be accurate.
No home	The scanner was unable to locate the home position mark. Scanning may proceed, but the document may not be scanned properly.
Shade error	Three unsuccessful attempts were made to calibrate the charge-coupled device (CCD) array correctly. Scanning may continue, but the results may be poor. The cause may be a dim lamp or a component that needs servicing.

Optical gain

Indicates the relative intensity of the lamp by returning the gain by which the optical system must boost the analog amplifier to give correct shading results. (This condition is not an error and does not set the Sense key).

REQUEST SENSE Return Structure Description for the OneScanner

The REQUEST SENSE command returns a structure containing the requested information. Figure 7-4 shows the format of the return structure for the OneScanner.

Figure 7-4 The REQUEST SENSE return structure for the OneScanner

Bit / Byte	7	6	5	4	3	2	1	0
0	($00) Reserved	(7)			($00) Reserved			
1	($00) Reserved							
2	($00) Reserved				Sense key			
3 - 6	($00) Reserved							
7	($0E) Additional sense length							
8 - 17	($00) Reserved							
18	Dim light	No light	No home	No limit	CCDCG	LRAM	CRAM	ROM
19	($00) Reserved				SRAM	CPU	($00) Reserved	
20	Optical gain							
21	($00) Reserved							

Here is a list of the fields and bits defined in this return structure:

Reserved — These fields are reserved for future expansion. Set them to 0.

Sense key — Bits 0–3 of byte 2 contain the Sense key. The possible Sense key values are as follows:

- $0 — No sense information: there is no specific sense data to be returned to the host computer. This situation occurs after a command executes successfully.
- $2 — Not ready: the scanner cannot scan without some user intervention.
- $4 — Hardware error: the scanner detects a nonrecoverable hardware fault while executing a command or during a self-test; these errors are indicated by the setting of the hardware error flags in bytes 18 and 19 of the return data block.
- $5 — Illegal request: the scanner detects an illegal parameter in the command block; the scanner cannot perform a scan and cannot alter any parameter data within the scanner.
- $6 — Unit attention: the scanner has just been switched on or reset and cannot execute any command except REQUEST SENSE or INQUIRY.
- $9 — Vendor unique: the scanner detects a vendor unique error, which is identified by the setting of the Vendor unique error flags in byte 18 of the return data block.

Additional sense length

This field indicates the number of sense data bytes that follow. The value of this field does not include bytes 0–7 of the return structure. It reflects the maximum size possible for a returned structure. The scanner does not adjust the value of this field to accommodate the Transfer length parameter specified in your REQUEST SENSE command.

Hardware error flags

Bytes 18 and 19 in the return structure contain bit flags that indicate the specific nature of a hardware error. Flags that are set to 1 indicate a specific hardware error condition. If the returned Sense key field is set to $4, then one or more of these bits are set to 1.

- No light — No light is detected from the fluorescent lamp.
- No limit — The scanner was unable to locate the limit switch position. Scanning cannot proceed because physical damage to the device may occur.
- CCDCG — CCD clock generator is not functioning properly. Unit should be repaired.
- LRAM — Line RAM has one or more bad bits. Unit should be repaired.
- CRAM — Correction RAM has one or more bad bits. Unit should be repaired.
- ROM — Bad checksum in the control ROM. Unit should be repaired.
- SRAM — General-purpose RAM for the scanner CPU has one or more bad bits. Unit should be repaired.
- CPU — CPU is not functioning properly. Unit should be repaired.

Vendor unique error flag

Bit 7 of byte 18 in the return structure contains a bit flag that indicates the specific nature of a vendor unique error, in this case, Dim light. If the flag is set, it indicates the specific error condition.

- Dim light — The fluorescent lamp is functioning, but its output has dropped below 70 percent and is too low for a scan to be accurate.

Optical gain

Indicates the relative intensity of the lamp by returning the gain by which the optical system must boost the analog amplifier to give correct shading results. (This condition is not an error and does not set the Sense key).

REQUEST SENSE Return Structure Description for the Color OneScanner

The REQUEST SENSE command returns a structure containing the requested information. Figure 7-5 shows the format of the return structure for the Color OneScanner.

Figure 7-5 The REQUEST SENSE return structure for the Color OneScanner

<table>
<tr><th>Bit
Byte</th><th>7</th><th>6</th><th>5</th><th>4</th><th>3</th><th>2</th><th>1</th><th>0</th></tr>
<tr><td>0</td><td>($00) Reserved</td><td colspan="3">($70) Current error</td><td colspan="4">($00) Reserved</td></tr>
<tr><td>1</td><td colspan="8">($00) Reserved</td></tr>
<tr><td>2</td><td colspan="4">($00) Reserved</td><td colspan="4">Sense key</td></tr>
<tr><td>3 - 6</td><td colspan="8">($00) Reserved</td></tr>
<tr><td>7</td><td colspan="8">($0E) Additional sense length</td></tr>
<tr><td>8 - 17</td><td colspan="8">($00) Reserved</td></tr>
<tr><td>18</td><td>Dim light</td><td>No light</td><td>No home</td><td>No limit</td><td>Cal Ckt</td><td>PSRAM</td><td>SRAM</td><td>ROM</td></tr>
<tr><td>19</td><td colspan="3">($00) Reserved</td><td>ICP</td><td colspan="2">($00) Reserved</td><td>Over light</td><td>($00) Reserved</td></tr>
<tr><td>20</td><td colspan="8">Optical gain</td></tr>
<tr><td>21</td><td colspan="8">($00) Reserved</td></tr>
</table>

Here is a list of the fields and bits defined in this return structure:

Reserved — These fields are reserved for future expansion. Set them to 0.

Current error

This field indicates a current scanner error.

Sense key — Bits 0–3 of byte 2 contain the Sense key. The possible Sense key values are as follows:

- $0 — No sense information: there is no specific sense data to be returned to the host computer. This situation occurs after a command executes successfully.
- $2 — Not ready: the scanner cannot scan without some user intervention.
- $4 — Hardware error: the scanner detects a nonrecoverable hardware fault while executing a command or during a self-test; these errors are indicated by the setting of the hardware error flags in bytes 18 and 19 of the return data block.
- $5 — Illegal request: the scanner detects an illegal parameter in the command block; the scanner cannot perform a scan and cannot alter any parameter data within the scanner.
- $6 — Unit attention: the scanner has just been switched on or reset and cannot execute any command except REQUEST SENSE or INQUIRY.
- $9 — Vendor unique: the scanner detects a vendor unique error, which is identified by the setting of the Vendor unique error flags in byte 18 of the return data block.

Additional sense length

This field indicates the number of sense data bytes that follow. The value of this field does not include bytes 0–7 of the return structure. It reflects the maximum size possible for a returned structure. The scanner does not adjust the value of this field to accommodate the Transfer length parameter specified in your REQUEST SENSE command.

Hardware error flags

Bytes 18 and 19 in the return structure contain bit flags that indicate the specific nature of a hardware error. Flags that are set to 1 indicate a specific hardware error condition. If the returned Sense key field is set to $4, then one or more of these bits are set to 1.

- No light — No light is detected from the fluorescent lamp.
- No limit — The scanner was unable to locate the limit switch position. Scanning cannot proceed because physical damage to the device may occur.
- Cal Ckt — The calibration circuit cannot support normal shading correction. Scanning should not proceed.
- PSRAM — The correction RAM has one or more bad bits. Scanning should not proceed.
- SRAM — General-purpose RAM for the scanner CPU has one or more bad bits. Unit should be repaired.
- ROM — Bad checksum in the control ROM. Unit should be repaired.
- ICP — The ICP (image correction processor) is not functioning properly after a read/write test of the ICP registers. Sense key $4 should be set when the ICP error bit is set to 1. Scanning should not proceed.CPU is not functioning properly. Unit should be repaired.

Vendor unique error flag

Byte 18 in the return structure contains bit flags that indicate the specific nature of a vendor unique error. Flags that are set to 1 indicate a specific error condition. If the returned Sense key field is set to $9, then one or more of these bits is set to 1.

Dim light	The fluorescent lamp is functioning, but its output has dropped below 70 percent and is too low for a scan to be accurate. Lamp needs replacing. The white target underneath the scanning bed is dirty. (Color OneScanner only) The three mirrors are dirty. (Color OneScanner only) The aperture plate is dirty. (Color OneScanner only) The CCD is dirty or bad. (Color OneScanner only) Scanning may continue, but results may not be satisfactory.
No home	The scanner was unable to locate the home position mark. Scanning may proceed, but the document may not be scanned properly.
Over light	The fluorescent lamp is too bright for normal scanning. Scanning may proceed, but the document may not be scanned properly.

Optical gain

Indicates the relative intensity of the lamp by returning the gain by which the optical system must boost the analog amplifier to give correct shading results. (This condition is not an error and does not set the Sense key).

INQUIRY ($12)

The INQUIRY command returns information about the scanner's device type, firmware version, conformance to various industry standards, and support for the SCSI command set. The following figure shows the format of the INQUIRY command structure. The scanner returns the requested information in a return structure appropriate to the scanner type. These return structures are described later in this section.

Bit Byte	7	6	5	4	3	2	1	0
0	($12) Operation code							
1	Logical unit number			($00) Reserved				EVPD
2	Page code							
3	($00) Reserved							
4	Allocation length							
5	Control							

An enable vital product data (EVPD) bit of one specifies that the target shall return the optional vital product data specified by the page code field. If the target does not support vital product data and this bit is set to one, the target shall return CHECK CONDITION status with the sense key set to ILLEGAL REQUEST and an additional sense code of INVALID FIELD IN CDB.
An EVPD bit of zero specifies that the target shall return the standard INQUIRY data. If the page code field is not zero, the target shall return CHECK CONDITION status with the sense key set to ILLEGAL REQUEST and an additional sense code of INVALID FIELD IN CDB.

The page code field specifies which page of vital product data information the target shall return (see 8.3.4).

The INQUIRY command shall return CHECK CONDITION status only when the target cannot return the requested INQUIRY data.

NOTE 64 The INQUIRY data should be returned even though the peripheral device may not be ready for other commands.
If an INQUIRY command is received from an initiator with a pending unit attention condition (i.e. before the target reports CHECK CONDITION status), the target shall perform the INQUIRY command and shall not clear the unit attention condition (see 7.9).

NOTES
65 The INQUIRY command is typically used by the initiator after a reset or power-up condition to determine the device types for system configuration. To minimize delays after a reset or power-up condition, the standard INQUIRY data should be available without incurring any media access delays. If the target does store some of the INQUIRY data on the device, it may return zeros or ASCII spaces (20h) in those fields until the data is available from the device.
66 The INQUIRY data may change as the target executes its initialization sequence or in response to a CHANGE DEFINITION command. For example, the target may contain a minimum command set in its non-volatile memory and may load its final firmware from the device when it becomes ready. After it has loaded the firmware, it may support more options and therefore return different supported options information in the INQUIRY data.

Standard INQUIRY data

The standard INQUIRY data (see table 45) contains 36 required bytes, followed by a variable number of vendor-specific parameters. Bytes 56 through 95, if returned, are reserved for future standardization.

Table 45 - Standard INQUIRY data format

Bit Byte	7	6	5	4	3	2	1	0
0	Peripheral qualifier			Peripheral device type				
1	RMB	Device-type modifier						
2	ISO version		ECMA version			ANSI-approved version		
3	AENC	TrmIOP	Reserved		Response data format			
4	Additional length							
5 - 6	Reserved							
7	RelAdr	WBus32	WBus16	Sync	Linked	Reserved	CmdQue	SftRe
8 - 17	Vendor identification							
16 - 31	Product identification							
32 - 35	Product revision level							

The peripheral qualifier and peripheral device type fields identify the device currently connected to the logical unit. If the target is not capable of supporting a device on this logical unit, this field shall be set to 7Fh (peripheral qualifier set to 011b and peripheral device type set to 1Fh). The peripheral qualifier is defined in table 46 and the peripheral device type is defined in table 47.

Table 46 - Peripheral qualifier

Code	Description
000b	The specified peripheral device type is currently connected to this logical unit. If the target cannot determine whether or not a physical device is currently connected, it shall also use this peripheral qualifier when returning the INQUIRY data. This peripheral qualifier does not mean that the device is ready for access by the initiator.
001b	The target is capable of supporting the specified peripheral device type on this logical unit; however, the physical device is not currently connected to this logical unit.
010b	Reserved
011b	The target is not capable of supporting a physical device on this logical unit. For this peripheral qualifier the peripheral device type shall be set to 1Fh to provide compatibility with previous versions of SCSI. All other peripheral device type values are reserved for this peripheral qualifier.
1XXb	Vendor-specific

Table 47 - Peripheral device type

Code	Description
00h	Direct-access device (e.g. magnetic disk)
01h	Sequential-access device (e.g. magnetic tape)
02h	Printer device
03h	Processor device
04h	Write-once device (e.g. some optical disks)
05h	CD-ROM device
06h	Scanner device
07h	Optical memory device (e.g. some optical disks)
08h	Medium changer device (e.g. jukeboxes)
09h	Communications device
0Ah - 0Bh	Defined by ASC IT8 (Graphic arts pre-press devices)
0Ch - 1Eh	Reserved
1Fh	Unknown of no device type

A removable medium (RMB) bit of zero indicates that the medium is not removable. A RMB bit of one indicates that the medium is removable.

The device-type modifier field was defined in SCSI-1 to permit vendor- specific qualification codes of the device type. This field is retained for compatibility with SCSI-1. Targets that do not support this field should return a value of zero.

The usage of non-zero code values in the ISO version and ECMA version fields are defined by the International Organization for Standardization and the European Computer Manufacturers Association, respectively. A zero code value in these fields shall indicate that the target does not claim compliance to the ISO version of SCSI (ISO 9316) or the ECMA version of SCSI (ECMA-111). It is possible to claim compliance to more than one of these SCSI standards.

The ANSI-approved version field indicates the implemented version of this International Standard and is defined in table 48.

Table 48 - ANSI-approved version

Code	Description
0h	The device might or might not comply to an ANSI-approved standard.
1h	The device complies to ANSI X3.131-1986 (SCSI-1).
2h	The device complies to this version of SCSI. This code is reserved to designate this standard upon approval by ANSI.
3h - 7h	Reserved

The asynchronous event notification capability (AENC) bit indicates that the device supports the asynchronous event notification capability as defined in 7.5.5.
a) Processor device-type definition: An AENC bit of one indicates that the processor device is capable of accepting asynchronous event notifications. An AENC bit of zero indicates that the processor device does not support asynchronous event notifications.
b) All other device-types: This bit is reserved.
A terminate I/O process (TrmIOP) bit of one indicates that the device supports the TERMINATE I/O PROCESS message as defined in 6.6.22. A value of zero indicates that the device does not support the TERMINATE I/O PROCESS message.

A response data format value of zero indicates the INQUIRY data format is as specified in SCSI-1. A response data format value of one indicates compatibility with some products that were designed prior to the development of this standard (i.e. CCS). A response data format value of two indicates that the data shall be in the format specified in this International Standard. Response data format values greater than two are reserved.

The additional length field shall specify the length in bytes of the parameters. If the allocation length of the command descriptor block is too small to transfer all of the parameters, the additional length shall not be adjusted to reflect the truncation.

A relative addressing (RelAdr) bit of one indicates that the device supports the relative addressing mode for this logical unit. If this bit is set to one, the linked command (Linked) bit shall also be set to one; since relative addressing can only be used with linked commands. A RelAdr bit of zero indicates the device does not support relative addressing for this logical unit.

A wide bus 32 (Wbus32) bit of one indicates that the device supports 32- bit wide data transfers. A value of zero indicates that the device does not support 32-bit wide data transfers.

A wide bus 16 (Wbus16) bit of one indicates that the device supports 16- bit wide data transfers. A value of zero indicates that the device does not support 16-bit wide data transfers.

NOTE 67 If the values of both the Wbus16 and Wbus32 bits are zero, the device only supports 8-bit wide data transfers.
A synchronous transfer (Sync) bit of one indicates that the device supports synchronous data transfer. A value of zero indicates the device does not support synchronous data transfer.

A linked command (Linked) bit of one indicates that the device supports linked commands for this logical unit. A value of zero indicates the device does not support linked commands for this logical unit.

A command queuing (CmdQue) bit of one indicates that the device supports tagged command queuing for this logical unit. A value of zero indicates the device does not support tagged command queuing for this logical unit.

A soft reset (SftRe) bit of zero indicates that the device responds to the RESET condition with the hard RESET alternative (see 6.2.2.1). A SftRe bit of one indicates that the device responds to the RESET condition with the soft RESET alternative (see 6.2.2.2).

ASCII data fields shall contain only graphic codes (i.e. code values 20h through 7Eh). Left-aligned fields shall place any unused bytes at the end of the field (highest offset) and the unused bytes shall be filled with space characters (20h). Right-aligned fields shall place any unused bytes at the start of the field (lowest offset) and the unused bytes shall be filled with space characters (20h).

The vendor identification field contains eight bytes of ASCII data identifying the vendor of the product. The data shall be left aligned within this field.

NOTE 68 It is intended that this field provide a unique vendor identification of the manufacturer of the SCSI device. In the absence of a formal registration procedure, X3T9.2 maintains a list of vendor identification codes in use. Vendors are requested to voluntarily submit their identification codes to X3T9.2 to prevent duplication of codes (see annex E).
The product identification field contains sixteen bytes of ASCII data as defined by the vendor. The data shall be left-aligned within this field.

The product revision level field contains four bytes of ASCII data as defined by the vendor. The data shall be left-aligned within this field.

Vital product data

Implementation of vital product data is optional. The information returned consists of configuration data (e.g. vendor identification, product identification, model, serial number), manufacturing data (e.g. plant and date of manufacture), field replaceable unit data and other vendor- or device-specific data.

The initiator requests the vital product data information by setting the EVPD bit to one and specifying the page code of the desired vital product data (see 8.3.4). If the target does not implement the requested page it shall return CHECK CONDITION status. The a sense key shall be set to ILLEGAL REQUEST and the additional sense code shall be set to INVALID FIELD IN CDB.

NOTES
69 The target should have the ability to execute the INQUIRY command even when a device error occurs that prohibits normal command execution. In such a case, CHECK CONDITION status should be returned for commands other than INQUIRY or REQUEST SENSE. The sense data returned may contain the field replaceable unit code. The vital product data should be obtained for the failing device using the INQUIRY command.
70 This International Standard defines a format that allows device- independent initiator software to display the vital product data returned by the INQUIRY command. For example, the initiator may display the data associated for the field replaceable unit returned in the sense data. The contents of the data may be vendor-specific; therefore, it may not be usable without detailed information about the device.

71 This International Standard does not define the location or method of storing the vital product data. The retrieval of the data may require completion of initialization operations within the device that may induce delays before the data is available to the initiator. Time-critical requirements are an implementation consideration and are not addressed in this International Standard.

INQUIRY Return Structure Description for the Apple Scanner

The INQUIRY return structure for the Apple Scanner contains a 5-byte header followed by 44 bytes of additional inquiry information, as shown in Figure 7-7.

Figure 7-7 The INQUIRY return structure for the Apple Scanner

Bit / Byte	7	6	5	4	3	2	1	0
0	($06) Device type							
1	($00) Reserved							
2	($02) Version							
3	($0) Reserved				($2) Response data format			
4	($2C) Additional length							
5 - 7	($00) Reserved							
8 - 17	Vendor identification							
16 - 31	Product identification							
32 - 35	Revision level							
36	($00) Reserved							
37	($00) Reserved							
38	($00) Group 0 commands opcode							
39	($90) Enabled opcodes: $00, $03							
40	($00) Enabled opcodes: none							
41	($27) Enabled opcodes: $12, $15, $16, $17							
42	($34) Enabled opcodes: $1A, $1B, $1D							
43	($01) Groupe 1 commands opcode							
44	($08) Enabled opcodes: $24							
45	($A0) Enabled opcodes: $28, $2A							
46	($08) Enabled opcodes: $34							
47	($00) Enabled opcodes: none							
48	($FF) End of block							

Here is a list of the fields defined in this return structure:

Device type — This field contains the SCSI standard device code of the scanner, which is $06.

Reserved — These fields are reserved for future expansion. Set them to 0.

Version — This field indicates the version number of this scanner, which is $02, as specified in the ANSI SCSI-2 (X3T9.2/86-109) device interface specification.

Response data format
: This field indicates that the inquiry data format is as specified in the ANSI SCSI-2 (X3T9.2/86-109) specification. The value of this field is $02 for the Color OneScanner.

Additional length
: This field indicates the number of data bytes that follow. The value of this field is $2C and does not include bytes 0–4 of the return structure. The scanner does not adjust the value of this field to accommodate the Transfer length specified in your INQUIRY command.

Vendor identification
: This field contains an ASCII string that identifies the scanner’s vendor. For the Color OneScanner, this string is "APPLE", followed by three spaces.

Product identification
: This field contains an ASCII string identifying the scanner’s product name and a model number. For the Color OneScanner, this string is "SCANNER A9M0337", followed by one space.

Revision level
: This field contains an ASCII string that identifies the scanner’s firmware revision level. For the Apple Scanner, this string is "0.00".

Group 0 commands
A 0 in this field indicates that at least one of the Group 0 scanning command operation codes (as defined in the ANSI SCSI-2 specification) is supported. Bytes 39–42 contain bit flags indicating the Group 0 commands supported by the Apple Scanner. Each command has a corresponding bit in the appropriate bit flag. That bit is set to 1 if the scanner supports the command, and it is set to 0 if the scanner does not support the command.

Enabled opcodes ($00 through $07)
These bit flags indicate the SCSI commands from $00 to $07 that the Apple Scanner supports. The Apple Scanner supports commands $00 (TEST UNIT READY) and $03 (INQUIRY). Therefore, bits 7 and 4 are set to 1. All other bits are set to 0. The value of the field is $90.

Enabled opcodes ($08 through $0F)
None of these operation codes is supported; all bits are set to 0.

Enabled opcodes ($10 through $17)
These bit flags indicate the SCSI commands from $10 to $17 that the Apple Scanner supports. The Apple Scanner supports commands $12 (INQUIRY), $15 (MODE SELECT), $16 (RESERVE), and $17 (RELEASE). Therefore, bits 5, 2, 1, and 0 are set to 1. All other bits are set to 0. The value of the field is $27.

Enabled opcodes ($18 through $1F)
These bit flags indicate the SCSI commands from $18 to $1F that the Apple Scanner supports. The Apple Scanner supports commands $1A (MODE SENSE), $1B (SCAN), and $1D (SEND DIAGNOSTIC). Therefore, bits 5, 4, and 2 are set to 1. All other bits are set to 0. The value of the field is $34.

Group 1 commands
A value of 1 in this field indicates that at least one of the Group 1 scanning commands (as defined in the ANSI SCSI-2 specification) is supported. Bytes 44–47 contain bit flags indicating the Group 1 commands supported by the Apple Scanner. Each command has a corresponding bit in the appropriate bit flag. That bit is set to 1 if the scanner supports the command, and it is set to 0 if the Color scanner does not support the command.

Enabled opcodes ($20 through $27)
These bit flags indicate the SCSI commands from $20 to $27 that the Apple Scanner supports. The Apple Scanner supports command $24 (DEFINE WINDOW PARAMETERS). Therefore, bit 3 is set to 1. All other bits are set to 0. The value of the field is $08.

Enabled opcodes ($28 through $2F)
These bit flags indicate the SCSI commands from $28 to $2F that the Apple Scanner supports. The Apple Scanner supports commands $28 (READ) and $2A (SEND). Therefore, bits 7 and 5 are set to 1. All other bits are set to 0. The value of the field is $A0.

Enabled opcodes ($30 through $37)
These bit flags indicate the SCSI commands from $30 to $37 that the Apple Scanner supports. The Apple Scanner supports command $34 (GET DATA STATUS). Therefore, bit 3 is set to 1. All other bits are set to 0. The value of the field is $08.

Enabled opcodes ($38 through $3F)
None of these operation codes is supported; all bits are set to 0.

End of block A value of $FF in this field is used to indicate the end of the block.

INQUIRY Return Structure Description for the OneScanner

The INQUIRY return structure for the OneScanner contains a 5-byte header followed by 44 bytes of additional inquiry information, as shown in Figure 7-8.

Figure 7-8 The INQUIRY return structure for the OneScanner

Bit / Byte	7	6	5	4	3	2	1	0
0	($06) Device type							
1	($00) Reserved							
2	($02) Version							
3	($0) Reserved				($2) Response data format			
4	($2C) Additional length							
5	($20) ROM size							
6	($04) CRAM size							
7	($08) SRAM size							
8 - 17	Vendor identification							
16 - 31	Product identification							
32 - 35	Revision level							
36	($00) Buffer space (MSB)							
37	($20) Buffer space (LSB)							
38	($00) Group 0 commands opcode							
39	($90) Enabled opcodes: $00, $03							
40	($00) Enabled opcodes: none							
41	($27) Enabled opcodes: $12, $15, $16, $17							
42	($34) Enabled opcodes: $1A, $1B, $1D							
43	($01) Groupe 1 commands opcode							
44	($08) Enabled opcodes: $24							
45	($A0) Enabled opcodes: $28, $2A							
46	($48) Enabled opcodes: $31, $34							
47	($00) Enabled opcodes: none							
48	($FF) End of block							

Here is a list of the fields defined in this return structure:

Device type — This field contains the SCSI standard device code of the scanner, which is $06.

Reserved — These fields are reserved for future expansion. Set them to 0.

Version — This field indicates the version number of this scanner, which is $02, as specified in the ANSI SCSI-2 (X3T9.2/86-109) device interface specification.

Response data format
: This field indicates that the inquiry data format is as specified in the ANSI SCSI-2 (X3T9.2/86-109) specification. The value of this field is $02 for the OneScanner.

Additional length
: This field indicates the number of data bytes that follow. The value of this field is $2C and does not include bytes 0–4 of the return structure. The scanner does not adjust the value of this field to accommodate the Transfer length specified in your INQUIRY command.

ROM size — This field indicates the size, in kilobytes, of the scanner ROM. The value of this field is $20.

CRAM size — This field indicates the size, in kilobytes, of the correction RAM. The value of this field is $04.

SRAM size — This field indicates the size, in kilobytes, of the general storage RAM for the scanner CPU. The value of this field is $08.

Vendor identification
: This field contains an ASCII string that identifies the scanner’s vendor. For the OneScanner, this string is "APPLE", followed by three spaces.

Product identification
: This field contains an ASCII string identifying the scanner’s product name and a model number. For the OneScanner, this string is "SCANNER II", followed by six spaces.

Revision level
: This field contains an ASCII string that identifies the scanner’s firmware revision level. For the Apple Scanner, this string is "2.02".

Buffer space — Bytes 36 and 37 indicate the total amount of buffer memory, in kilobytes, in the attached scanner. Buffer memory is used to store scan image data until the host computer reads the data from the scanner. Together, these two bytes contain a 16-bit value: byte 36 contains the most significant byte; byte 37 contains the least significant byte. The value of byte 36 is $00; the value of byte 37 is $20.

Group 0 commands
A 0 in this field indicates that at least one of the Group 0 scanning command operation codes (as defined in the ANSI SCSI-2 specification) is supported. Bytes 39–42 contain bit flags indicating the Group 0 commands supported by the OneScanner. Each command has a corresponding bit in the appropriate bit flag. That bit is set to 1 if the OneScanner supports the command, and it is set to 0 if the OneScanner does not support the command.

Enabled opcodes ($00 through $07)
These bit flags indicate the SCSI commands from $00 to $07 that the OneScanner supports. The OneScanner supports commands $00 (TEST UNIT READY) and $03 (INQUIRY). Therefore, bits 7 and 4 are set to 1. All other bits are set to 0. The value of the field is $90.

Enabled opcodes ($08 through $0F)
None of these operation codes is supported; all bits are set to 0.

Enabled opcodes ($10 through $17)
These bit flags indicate the SCSI commands from $10 to $17 that the OneScanner supports. The OneScanner supports commands $12 (INQUIRY), $15 (MODE SELECT), $16 (RESERVE), and $17 (RELEASE). Therefore, bits 5, 2, 1, and 0 are set to 1. All other bits are set to 0. The value of the field is $27.

Enabled opcodes ($18 through $1F)
These bit flags indicate the SCSI commands from $18 to $1F that the OneScanner supports. The OneScanner supports commands $1A (MODE SENSE), $1B (SCAN), and $1D (SEND DIAGNOSTIC). Therefore, bits 5, 4, and 2 are set to 1. All other bits are set to 0. The value of the field is $34.

Group 1 commands
A value of 1 in this field indicates that at least one of the Group 1 scanning commands (as defined in the ANSI SCSI-2 specification) is supported. Bytes 44–47 contain bit flags indicating the Group 1 commands supported by the OneScanner. Each command has a corresponding bit in the appropriate bit flag. That bit is set to 1 if the OneScanner supports the command, and it is set to 0 if the Onecanner does not support the command.

Enabled opcodes ($20 through $27)
These bit flags indicate the SCSI commands from $20 to $27 that the OneScanner supports. The OneScanner supports command $24 (DEFINE WINDOW PARAMETERS). Therefore, bit 3 is set to 1. All other bits are set to 0. The value of the field is $08.

Enabled opcodes ($28 through $2F)
These bit flags indicate the SCSI commands from $28 to $2F that the OneScanner supports. The OneScanner supports commands $28 (READ) and $2A (SEND). Therefore, bits 7 and 5 are set to 1. All other bits are set to 0. The value of the field is $A0.

Enabled opcodes ($30 through $37)
These bit flags indicate the SCSI commands from $30 to $37 that the OneScanner supports. The OneScanner supports command $31 (OBJECT POSITION) and $34 (GET DATA STATUS). Therefore, bits 6 and 3 are set to 1. All other bits are set to 0. The value of the field is $48.

Enabled opcodes ($38 through $3F)
None of these operation codes is supported; all bits are set to 0.

End of block A value of $FF in this field is used to indicate the end of the block.

INQUIRY Return Structure Description for the Color OneScanner

The INQUIRY return structure for the Color OneScanner contains a 5-byte header followed by 48 bytes of additional inquiry information, as shown in Figure 7-9A.

Figure 7-9A The INQUIRY return structure for the Color OneScanner

Bit / Byte	7	6	5	4	3	2	1	0
0	($06) Device type							
1	($00) Reserved							
2	($02) Version							
3	($0) Reserved				($2) Response data format			
4	($30) Additional length							
5 - 7	($00) Reserved							
8 - 17	Vendor identification							
16 - 31	Product identification							
32 - 35	Revision level							
36	($00) Buffer space (MSB)							
37	($80) Buffer space (LSB)							
38	($00) Group 0 commands opcode							
39	($90) Enabled opcodes: $00, $03							
40	($00) Enabled opcodes: none							
41	($27) Enabled opcodes: $12, $15, $16, $17							
42	($34) Enabled opcodes: $1A, $1B, $1D							
43	($01) Groupe 1 commands opcode							
44	($0C) Enabled opcodes: $24, $25							
45	($A0) Enabled opcodes: $28, $2A							
46	($48) Enabled opcodes: $31, $34							
47	($00) Enabled opcodes: none							
48	($FF) End of block							
49	($40) ROM size							
50	($01) PSRAM size (MSB)							
51	($00) PSRAM size (LSB)							
52	($08) SRAM size							

Here is a list of the fields defined in this return structure:

Device type This field contains the SCSI standard device code of the scanner, which is $06.

Reserved These fields are reserved for future expansion. Set them to 0.

Version This field indicates the version number of this scanner, which is $02, as specified in the ANSI SCSI-2 (X3T9.2/86-109) device interface specification.

Response data format
This field indicates that the inquiry data format is as specified in the ANSI SCSI-2 (X3T9.2/86-109) specification. The value of this field is $02 for the Color OneScanner.

Additional length
This field indicates the number of data bytes that follow. The value of this field is $30 and does not include bytes 0-4 of the return structure. The scanner does not adjust the value of this field to accommodate the Transfer length specified in your INQUIRY command.

Vendor identification
This field contains an ASCII string that identifies the scanner’s vendor. For the Color OneScanner, this string is "APPLE", followed by three spaces.

Product identification
This field contains an ASCII string identifying the scanner’s product name and a model number. For the Color OneScanner, this string is "SCANNER III", followed by five spaces.

Revision level
This field contains an ASCII string that identifies the scanner’s firmware revision level. For the Color OneScanner, this string is "3.00".

Buffer space Bytes 36 and 37 indicate the total amount of buffer memory, in kilobytes, in the attached scanner. Buffer memory is used to store scan image data until the host computer reads the data from the scanner. Together, these two bytes contain a 16 bit value: byte 36 contains the most significant byte; byte 37 contains the least significant byte. The value of byte 36 is $00; the value of byte 37 is $80.

Group 0 commands
A 0 in this field indicates that at least one of the Group 0 scanning command operation codes (as defined in the ANSI SCSI-2 specification) is supported. Bytes 39–52 contain bit flags indicating the Group 0 commands supported by the Color OneScanner. Each command has a corresponding bit in the appropriate bit flag. That bit is set to 1 if the Color OneScanner supports the command, and it is set to 0 if the Color OneScanner does not support the command.

Enabled opcodes ($00 through $07)
These bit flags indicate the SCSI commands from $00 to $07 that the Color OneScanner supports. The Color OneScanner supports commands $00 (TEST UNIT READY) and $03 (INQUIRY). Therefore, bits 7 and 4 are set to 1. All other bits are set to 0. The value of the field is $90.

Enabled opcodes ($08 through $0F)
None of these operation codes is supported; all bits are set to 0.

Enabled opcodes ($10 through $17)
These bit flags indicate the SCSI commands from $10 to $17 that the Color OneScanner supports. The Color OneScanner supports commands $12 (INQUIRY), $15 (MODE SELECT), $16 (RESERVE), and $17 (RELEASE). Therefore, bits 5, 2, 1, and 0 are set to 1. All other bits are set to 0. The value of the field is $27.

Enabled opcodes ($18 through $1F)
These bit flags indicate the SCSI commands from $18 to $1F that the Color OneScanner supports. The Color OneScanner supports commands $1A (MODE SENSE), $1B (SCAN), and $1D (SEND DIAGNOSTIC). Therefore, bits 5, 4, and 2 are set to 1. All other bits are set to 0. The value of the field is $34.

Group 1 commands
A value of 1 in this field indicates that at least one of the Group 1 scanning commands (as defined in the ANSI SCSI-2 specification) is supported. Bytes 44–47 contain bit flags indicating the Group 1 commands supported by the Color OneScanner. Each command has a corresponding bit in the appropriate bit flag. That bit is set to 1 if the Color OneScanner supports the command, and it is set to 0 if the Color OneScanner does not support the command.

Enabled opcodes ($20 through $27)
These bit flags indicate the SCSI commands from $20 to $27 that the Color OneScanner supports. The Color OneScanner supports commands $24 (DEFINE WINDOW PARAMETERS) and $25 (GET WINDOW PARAMETERS). Therefore, bits 2 and 3 are set to 1. All other bits are set to 0. The value of the field is $0C.

Enabled opcodes ($28 through $2F)
These bit flags indicate the SCSI commands from $28 to $2F that the Color OneScanner supports. The Color OneScanner supports commands $28 (READ) and $2A (SEND). Therefore, bits 7 and 5 are set to 1. All other bits are set to 0. The value of the field is $A0.

Enabled opcodes ($30 through $37)
These bit flags indicate the SCSI commands from $30 to $37 that the Color OneScanner supports. The Color OneScanner supports commands $31 (OBJECT POSITION) and $34 (GET DATA STATUS). Therefore, bits 6 and 3 are set to 1. All other bits are set to 0. The value of the field is $48.

Enabled opcodes ($38 through $3F)
None of these operation codes is supported; all bits are set to 0.

End of block — A value of $FF in this field is used to indicate the end of the block.

ROM size — This field indicates the size, in kilobytes, of the scanner ROM. The value of this field is $40, equivalent to 64K decimal.

PSRAM size — This field is a two-byte field that indicates the size, in kilobytes, of the correction RAM. It contains $01 and $00, which are equivalent to 256K decimal.

SRAM size — This field indicates the size, in kilobytes, of the general storage RAM for the scanner CPU. The value of this field is $08.

INQUIRY Return Structure Description for the Color OneScanner 600/27

The INQUIRY return structure for the Color OneScanner contains a 5-byte header followed by 31 bytes of additional inquiry information, as shown in Figure 7-9B.

Figure 7-9B The INQUIRY return structure for the Color OneScanner 600/27

Bit Byte	7	6	5	4	3	2	1	0
0	($06) Device type							
1	($00) Reserved							
2	($02) Version							
3	($0) Reserved				($2) Response data format			
4	($1F) Additional length							
5 - 6	($00) Reserved							
7	($10) Reserved							
8 - 17	Vendor identification							
16 - 31	Product identification							
32 - 35	Revision level							
36	($FF) End of block							

Here is a list of the fields defined in this return structure:

Device type This field contains the SCSI standard device code of the scanner, which is $06.

Reserved These fields are reserved for future expansion. Set them to 0.

Version This field indicates the version number of this scanner, which is $02, as specified in the ANSI SCSI-2 (X3T9.2/86-109) device interface specification.

Response data format
This field indicates that the inquiry data format is as specified in the ANSI SCSI-2 (X3T9.2/86-109) specification. The value of this field is $02 for the Color OneScanner 600/27.

Additional length
This field indicates the number of data bytes that follow. The value of this field is $1F and does not include bytes 0–4 of the return structure. The scanner does not adjust the value of this field to accommodate the Transfer length specified in your INQUIRY command.

Vendor identification
This field contains an ASCII string that identifies the scanner’s vendor. For the Color OneScanner 600/27, this string is "CANON", followed by three spaces.

Product identification
This field contains an ASCII string identifying the scanner’s product name and a model number. For the Color OneScanner 600/27, this string is "IX-03035B", followed by seven spaces.

Revision level
This field contains an ASCII string that identifies the scanner’s firmware revision level. For the Color OneScanner 600/27, this string is "1.01".

End of block A value of $FF in this field is used to indicate the end of the block.

INQUIRY Return Structure Description for the Color OneScanner 1200/30

The INQUIRY return structure for the Color OneScanner contains a 5-byte header followed by 31 bytes of additional inquiry information, as shown in Figure 7-9C.

Figure 7-9C The INQUIRY return structure for the Color OneScanner 1200/30

Bit Byte	7	6	5	4	3	2	1	0
0	($06) Device type							
1	($00) Reserved							
2	($02) Version							
3	($0) Reserved				($2) Response data format			
4	($1F) Additional length							
5 - 6	($00) Reserved							
7	($10) Reserved							
8 - 17	Vendor identification							
16 - 31	Product identification							
32 - 35	Revision level							
36	($FF) End of block							

Here is a list of the fields defined in this return structure:

Device type This field contains the SCSI standard device code of the scanner, which is $06.
Reserved These fields are reserved for future expansion. Set them to 0.
Version This field indicates the version number of this scanner, which is $02, as specified in the ANSI SCSI-2 (X3T9.2/86-109) device interface specification.

Response data format
: This field indicates that the inquiry data format is as specified in the ANSI SCSI-2 (X3T9.2/86-109) specification. The value of this field is $02 for the Color OneScanner 1200/30.

Additional length
: This field indicates the number of data bytes that follow. The value of this field is $1F and does not include bytes 0–4 of the return structure. The scanner does not adjust the value of this field to accommodate the Transfer length specified in your INQUIRY command.

Vendor identification
: This field contains an ASCII string that identifies the scanner’s vendor. For the Color OneScanner, this string is "CANON", followed by three spaces.

Product identification
: This field contains an ASCII string identifying the scanner’s product name and a model number. For the Color OneScanner 1200/30, this string is "IX-06015C", followed by seven spaces.

Revision level
: This field contains an ASCII string that identifies the scanner’s firmware revision level. For the Color OneScanner 1200/30, this string is "1.07".

End of block A value of $FF in this field is used to indicate the end of the block.

MODE SELECT ($15)

The MODE SELECT command passes a parameter block to the scanner. The scanner uses this data to configure its operating parameters. Figure 7-10 shows the format of the MODE SELECT command structure. The information is passed to the scanner in a MODE SELECT parameter list that is appropriate to the parameter type and attached scanner. These parameter lists are described later in this section.

Figure 7-10 The MODE SELECT command structure

Bit Byte	7	6	5	4	3	2	1	0
0	($15) Operation code							
1	Logical unit number			PF	($00) Reserved			SP
2	($00) Reserved							
3	($00) Reserved							
4	Parameter list length							
5	Control							

Here is a list of the fields and bits used in this command structure:

Operation code
: The operation code for this command is $15.

Reserved
: These fields are reserved for future expansion. Set them to 0.

Page format (PF)
: The Page format bit indicates that all parameters after the block descriptors are unique to the vendor, as specified in this reference. This bit must always be set to 1.

Save pages (SP)
: A save pages (SP) bit of zero indicates the target shall perform the specified MODE SELECT operation, and shall not save any pages. An SP bit of one indicates that the target shall perform the specified MODE SELECT operation, and shall save to a non-volatile vendor-specific location all the savable pages including any sent during the DATA OUT phase.

Parameter list length
: This field contains the length, in bytes, of the parameter list to be passed to the scanner. A transfer length of 0 indicates that no parameter list is being passed (this is not an error condition).

MODE SELECT Parameter List Description

The MODE SELECT parameter list contains one or more optional parameter pages. The parameter page may be either an Apple-specific parameter page or a disconnect-reconnect parameter page. Each parameter page includes 2 bytes that specify the type and the length of the data page. The data pages are described next.

Apple-Specific Data Page Description for the Apple Scanner

Figure 7-11 shows the Apple-specific data page for the Apple Scanner.
Figure 7-11 The MODE SELECT Apple-specific data page for the Apple Scanner

Bit Byte	7	6	5	4	3	2	1	0
0	($00) Reserved							
1	($00) Reserved							
2	($00) Reserved							
3	($00) Reserved							
4	($00) Page code							
5	($06) Page length							
6	($00) Reserved						Graymap	
7	Auto background-adjustment threshold							
8	($00) Reserved							Lamp
9	($00) Reserved							
10	($00) Reserved							
11	($00) Reserved							

Here is a list of the fields and bits used in this page:

Reserved These fields are reserved for future expansion. They are set to 0.

Page code This field identifies the page type. Set them to 0.

Page length This field contains the length, in bytes, of the data portion of the parameter page, not including bytes 0-5. This field is set to 6.

Graymap This field contains the value that controls the graymap function of the scanner. The graymap function allows the scanner to enhance detail in light areas or dark areas of a document, or not to enhance either. A value of 0 results in more detail in the darker areas of the document. A value of 1 results in no alteration of the data. A value of 2 results in more detail in the lighter areas of the document.

Auto background-adjustment threshold
This field indicates the threshold level at which the scanner determines whether a dot is black or white when the user selects automatic back- ground adjustment. The default threshold level is 64.

When excessively dark areas of an original document are scanned, automatic background adjustment constantly adjusts the brightness level, resulting in more detail in the darker areas. Light areas are left unaltered. To select automatic background adjustment, set the Threshold parameter to 0 in the DEFINE WINDOW PARAMETERS command parameter list. See “DEFINE WINDOW PARAMETERS ($24)” later in this chapter.

Lamp This bit controls the fluorescent lamp used to illuminate the document during scanning. A value of 1 turns on the lamp. A value of 0 turns it off.

In addition, other conditions may affect whether the lamp is on or off. Once the lamp is turned on, it remains on until two minutes elapse without SCSI activity. The next SCSI command turns on the lamp for a minimum period of two minutes. Furthermore, if the No home bit is set to 1 in the SCAN command, the scanner turns off the lamp only after the carriage assembly returns to the home position. See “SCAN ($1B),” later in this chapter.

Apple-Specific Data Page Description for the OneScanner

Figure 7-12 shows the Apple-specific data page for the OneScanner.
Figure 7-12 The MODE SELECT Apple-specific data page for the OneScanner

Bit / Byte	7	6	5	4	3	2	1	0
0	($00) Reserved							
1 - 3	($00) Reserved							
4	($00) Page code							
5	($06) Page length							
6	($00) Reserved							
7	($00) Reserved					LED	Reset	Button
8	($00) Reserved				Fast H	Fast L	CCD	Lamp
9	($00) Reserved							
10	($00) Reserved							
11	($00) Reserved							

Here is a list of the fields and bits used in this page:

Reserved — These fields are reserved for future expansion. They are set to 0.

Page code — This field identifies the page type. Set it to 0.

Page length — This field contains the length, in bytes, of the data portion of the parameter page, not including bytes 0–5. This field is set to 6.

LED — This bit indicates the setting of the amber LED on the OneScanner. A value of 1 indicates that the LED is on. A value of 0 indicates that the LED is off.

Reset — This bit indicates whether the scanner should reset the Button bit. A value of 1 causes the scanner to set the Button bit to 0. A value of 0 has no effect.

Button — This bit indicates the current state of the button on the OneScanner since the last reset request (a MODE SELECT command with the Reset bit set to 1). A value of 1 indicates that the button has been pressed. A value of 0 indicates that the button has not been pressed. Applies only to scanners with ROM version 2.03, or earlier.

Fast H, Fast L

Together, these bits indicate the scan speed for the scanner. Here are four possible values for these bits:

Fast H	Fast L	Scan speed
0	0	Normal speed
0	1	High speed
1	0	Fast speed
1	1	Invalid value

At normal speed, the scanner processes the image at the slowest possible carriage speed and performs handshaking on all data exchanges with the host computer. At high speed, the scanner runs the carriage at high speed but still performs handshaking on all data exchanges with the host computer. At normal and high speeds, data handshaking prevents the loss of image data.

At fast speed, the scanner runs the carriage at high speed but does not perform any handshaking during data exchanges with the host. Eliminating the handshaking greatly improves the data exchange rate between the scanner and the host computer. However, image data may be lost if the host computer cannot keep up with the scanner. If the scanner loses data due to an overrun condition, it aborts the scan with a CHECK CONDITION status.

CCD — This bit controls power to the CCD array in the scanner. A value of 1 causes the scanner to power on the CCD array whenever the scanner lamp is turned on. A value of 0 causes the scanner to power on the CCD array only in response to SCAN commands.

Lamp — This bit indicates the current state of the fluorescent lamp used to illuminate the document during scanning. A value of 1 indicates that the lamp is on; a value of 0 indicates that the lamp is off.

In addition, other conditions may affect whether the lamp is on or off. Once the lamp has been turned on, it remains on until two minutes elapse without SCSI activity. The next SCSI command turns on the lamp for a minimum period of two minutes.

Apple-Specific Data Page Description for the Color OneScanner

Figure 7-13 shows the Apple-specific data page for the Color OneScanner.

Figure 7-13 The MODE SENSE Apple-specific data page for the Color OneScanner

Bit / Byte	7	6	5	4	3	2	1	0
0	($00) Reserved							
1 - 3	($00) Reserved							
4	($00) Reserved		($00) Apple unique control parameters					
5	($06) Parameter length							
6	($00) Reserved							
7	($00) Reserved					Scan LED	Gamma reset	3x3 reset
8	($00) Reserved			MTF circuit	ICP bypass	Data polarity	CCD on	Lamp on
9	Code sense select							
10	($00) Reserved							
11	($00) Reserved							

Here is a list of the fields and bits used in this page:

Reserved These fields are reserved for future expansion. They are set to 0.

Apple-unique control parameters
Apple unique control parameters are reserved for specific Apple functions. This field is reserved for Page Code, which identifies the page type. Set it to 0.

Page code This field identifies the page type. Set it to 0.

Parameter length
This field contains the length, in bytes, of the data portion of the parameter page, not including bytes 0–5. This field is set to 6.

Scan LED This bit controls the setting of the amber LED on the Color OneScanner. A value of 1 turns on the LED. A value of 0 turns it off.

Gamma reset When this bit is set (1) the scanner loads the gamma table with default values, which are chosen so that the output data is the same as the input data. The default value is 0.

3x3 reset This bit directs the scanner to load the default 3-by-3 color correction table. When this bit is set (1), the following default 3-by-3 matrix is loaded. The default value is 0.

```
1      0      0
0      1      0
0      0      1
```

MTF circuit This bit indicates the setting of the MTF circuit.

1 MTF peaking circuit on
0 MTF peaking circuit off

ICP bypass This bit decides whether the ICP is to be used or bypassed. When the field is set (1) the circuit is bypassed. When it is reset (0) the circuit is used.

Data polarity
This bit determines the polarity of the output image data. When it is set (1) 00 is white and FF is black. In normal mode, it is reset to 0, and 00 is black and FF white.

CCD on This bit controls power to the CCD array in the scanner. A value of 1 causes the scanner to power on the CCD array whenever the scanner lamp is turned on. A value of 0 causes the scanner to power on the CCD array only in response to SCAN operations.

Lamp on This bit controls the fluorescent lamp used to illuminate the document during scanning. A value of 1 turns on the lamp. A value of 0 turns it off.
In addition, other conditions may affect whether the lamp is on or off. Once the lamp has been turned on, it remains on until two minutes elapse without SCSI activity. The next SCSI command turns on the lamp for a minimum period of two minutes.

Color sense select
This field indicates which color sensor line or lines are selected for scanning. The green line sensor is the default for grayscale scanning.

Disconnect-Reconnect Parameter Page Description

Figure 7-14 shows the disconnect-reconnect parameter page. You may use this page for the Apple Scanner, the OneScanner, and the Color OneScanner.

Here is a list of the fields used in this page:

Reserved	These fields are reserved for future expansion. Set them to 0.
Page code	This field identifies the parameter page as a disconnect-reconnect page. Set it to $02.
Page length	This field contains the length, in bytes, of the data portion of the parameter page, not including bytes 0–5. Set it to $0A.

Figure 7-14 The MODE SELECT disconnect-reconnect parameter page

Bit Byte	7	6	5	4	3	2	1	0
0 - 3	($00) Reserved							
4	($02) Page code							
5	($0A) Page length							
6	Buffer full ratio							
7 - 15	($00) Reserved							

Buffer full ratio

Indicates to the scanner how full the scanner’s image buffer must be before the scanner returns image data to the host computer in response to a GET DATA STATUS command (with the Wait bit set to 1). This field provides the numerator of a fraction whose denominator is 255. The resulting ratio controls how full the scanner buffer is allowed to get before data is returned to your program. The recommended value of 128 sets this threshold at 50 percent.

Measurement Units Parameter Page Description

Figure 7-14B shows the measurement units parameter page. You may use this page for the Color OneScanner 600/27 and the Color OneScanner 1200/30.

Here is a list of the fields used in this page:

Reserved These fields are reserved for future expansion. Set them to 0.

Page code This field identifies the parameter page as a measurement units page. Set it to $03.

Page length This field contains the length, in bytes, of the data portion of the parameter page, not including bytes 0-5. Set it to $06.

Figure 7-14B The MODE SELECT measurement units page

Bit Byte	7	6	5	4	3	2	1	0
0 - 3	($00) Reserved							
4	($03) Page code							
5	($06) Page length							
6	Basic measurement unit							
7	($00) Reserved							
8	Measurement unit divisor (MSB)							
9	Measurement unit divisor (LSB)							
10 - 11	($00) Reserved							

Basic measurement unit

The basic measurement unit field is defined in table below. Targets shall use inches as the default basic measurement unit.

Measurement unit divisor

The measurement unit divisor specifies the number of units needed to equal one basic measurement unit. Targets shall use 1200 as the default measurement unit divisor. If a value of zero is specified the target shall return CHECK CONDITION status and set the sense key to ILLEGAL REQUEST.

Note

A target that does not implement this page or only supports default values uses twelve hundredths (1/1200) of an inch as the unit of measure.

Basic measurement units

Code	Description
00h	Inch
01h	Millimètre
02h	Point
03h - FFh	Reserved

RESERVE UNIT ($16)

The RESERVE command prevents all devices other than the requesting device from accessing the scanner. The scanner rejects commands that it receives from devices other than the initiating host and instead returns a RESERVATION CONFLICT status. The host computer may reserve the target device for use by a third device by setting the third-party bit (3pty) in byte 1 of the command structure. In this case, only the third device can access the scanner to execute commands, although the initiating host computer may issue another RESERVE command or a RELEASE command. See "RELEASE ($17)," later in this chapter. Figure 7-15 shows the format of the RESERVE command structure.

The RESERVE command is terminated by the RELEASE command or by another RESERVE command issued from the original RESERVE command initiator.
The RESERVE and RELEASE commands together provide the basic mechanism for bus-contention resolution in a multidevice SCSI bus environment. The mechanism works as follows: an application running on a host computer must have uninterrupted access to the scanner while transferring commands and data to and from the scanner. The RESERVE command provides a means for an application to have access to the scanner until it is no longer needed. The application then issues a RELEASE command, which releases the scanner so that another SCSI device on the bus can use it.

Figure 7-15 The RESERVE command structure

Bit Byte	7	6	5	4	3	2	1	0
0	($16) Operation code							
1	Logical unit number			3rdPty	Third party device ID			($00) Reserved
2	($00) Reserved							
3	($00) Reserved							
4	($00) Reserved							
5	Control							

The RESERVE UNIT and RELEASE UNIT commands provide the basic mechanism for contention resolution in multiple-initiator systems.
This command requests that the entire logical unit be reserved for the exclusive use of the initiator until the reservation is superseded by another valid RESERVE UNIT command from the initiator that made the reservation or until released by a RELEASE UNIT command from the same initiator that made the reservation, by a BUS DEVICE RESET message from any initiator, by a hard reset condition, or by a power on cycle. The reservation shall not be granted if the logical unit is reserved by another initiator. It shall be permissible for an initiator to reserve a logical unit that is currently reserved by that initiator.

If the logical unit is reserved for another initiator, the target shall return RESERVATION CONFLICT status.

If, after honouring the reservation, any other initiator attempts to perform any command on the reserved logical unit other than an INQUIRY, REQUEST SENSE, PREVENT ALLOW MEDIUM REMOVAL (with a prevent bit of zero), or a RELEASE UNIT command, the command shall be rejected with RESERVATION CONFLICT status.

Third-party reservation

Third-party reservation allows an initiator to reserve a logical unit for another SCSI device. This is intended for use in multiple- initiator systems that use the COPY command.

If the third-party (3rdPty) bit is zero, a third-party reservation is not requested. If the 3rdPty bit is one the target shall reserve the logical unit for the SCSI device specified in the third-party device ID field. The target shall preserve the reservation until it is superseded by another valid RESERVE UNIT command from the initiator that made the reservation or until it is released by the same initiator, by a BUS DEVICE RESET message from any initiator, or a hard reset condition. The target shall ignore any attempt to release the reservation made by any other initiator.

If independent sets of parameters are implemented, a third party reservation shall cause the target to transfer the set of parameters in effect for the initiator of the RESERVE command to the parameters used for commands from the third-party device. Any subsequent command issued by the third-party device is executed according to the mode parameters in effect for the initiator that sent the RESERVE command.

If independent sets of parameters are implemented, a third party reservation shall cause the target to transfer the set of parameters in effect for the initiator of the RESERVE command to the parameters used for commands from the third party device. Any subsequent command issued by the third-party device is executed according to the mode parameters in effect for the initiator that sent the RESERVE command.

NOTE 147 This transfer of the mode parameters is applicable to target devices which store mode information independently for different initiators. This mechanism allows an initiator to set the mode parameters of a target for the use of a copy master (i.e. the third-party device). The third-party copy master may subsequently issue a MODE SELECT command to modify the mode parameters.

Superseding reservations

An initiator that currently has a logical unit reserved may modify the current reservation by issuing another RESERVE UNIT command to the same logical unit. The superseding reservation shall release the current reservation if the superseding reservation request is granted. The current reservation shall not be modified if the superseding reservation request cannot be granted. If the superseding reservation cannot be granted because of conflicts with a previous reservation (other than the current reservation), then the target shall return RESERVATION CONFLICT status.

NOTE 148 Superseding reservations allow the third-party SCSI device ID to be changed during a reservation using the third-party reservation option. This capability is necessary for certain situations when using COMPARE, COPY, and COPY AND VERIFY commands.

RELEASE UNIT ($17)

The RELEASE command removes the specified reservation on the scanner. Only the device that issued the original RESERVE command can execute RELEASE. See "RESERVE ($16)," earlier in this chapter, for more information. Figure 7-16 shows the format of the RELEASE command structure.

Figure 7-16 The RELEASE command structure

Bit / Byte	7	6	5	4	3	2	1	0
0	($17) Operation code							
1	Logical unit number			3rdPty	Third party device ID			($00) Reserved
2	($00) Reserved							
3	($00) Reserved							
4	($00) Reserved							
5	Control							

The RESERVE UNIT and RELEASE UNIT commands provide the basic mechanism for contention resolution in multiple-initiator systems.
If a valid reservation exists for the I_T_L nexus, the target shall release the reservation and return GOOD status.

A reservation may only be released by the initiator that made it. It is not an error to attempt to release a reservation that is not currently valid or is held by another initiator. In this case, the target shall return GOOD status without altering any other reservation.

Third-party release

Third-party release allows an initiator to release a logical unit that was previously reserved using a third-party reservation (see 10.2.10.1).

If the third party (3rdPty) bit is zero, then a third-party release is not requested. If the 3rdPty bit is one, and if the reservation was made using a third-party reservation by the initiator that is requesting the release for the same SCSI device as specified in the third-party device ID, then the target shall release the reservation.

If the 3rdPty bit is one, the target shall not modify the mode parameters for commands received from the third-party device even if the target implements the transfer of mode parameters with a third-party RESERVE UNIT command.

NOTE 146 When a target implements independent storage of mode parameters for each initiator, a third-party RESERVE UNIT command effects a transfer of the current mode parameters. Those set up by the initiator of the RESERVE UNIT are to be set as the mode parameters used for commands from the third-party device (usually a copy master device). A unit attention condition notifies the third-party device of the changed mode parameters. A successful third-party RELEASE UNIT command leaves the transferred parameters intact. The third-party device can issue MODE SENSE and MODE SELECT commands to query and modify the mode parameters.

MODE SENSE ($1A)

The MODE SENSE command allows the host computer to read back the parameter information that was sent to the scanner with the MODE SELECT command. See "MODE SELECT ($15)," earlier in this chapter, for more information. Figure 7-17 shows the format of the MODE SENSE command structure. The scanner returns the information to the host computer in one or more MODE SENSE data pages as appropriate for the request and the type of scanner attached. The format of the data pages is described later in this section.

Figure 7-17 The MODE SENSE command structure

Bit Byte	7	6	5	4	3	2	1	0
0	($1A) Operation code							
1	Logical unit number			PF	DBD	($00) Reserved		
2	PC		Page code					
3	($00) Reserved							
4	Parameter list length							
5	Control							

Here is a list of the fields and bits used in this command structure:

Operation code
The operation code for the MODE SENSE command is $1A.

Reserved These fields are reserved for future expansion. Set them to 0.

Page format (PF)
This bit indicates that the return data is formatted into pages. Set it to 1.

Disable Block Descriptors (DBD)
A disable block descriptors (DBD) bit of zero indicates that the target may return zero or more block descriptors in the returned MODE SENSE data (see 8.3.3), at the target's discretion. A DBD bit of one specifies that the target shall not return any block descriptors in the returned MODE SENSE data.

Page control (PC)
The page control (PC) field defines the type of mode parameter values to be returned in the mode pages. Here are the three possible values for these bits:
00b Current values
01b Changeable values
10b Default values
11b Saved values

Page code This field indicates which type of data page is to be returned.
Set it to 0 to request that the Apple-specific data page appropriate to the attached scanner be returned.
Set it to 2 to request that the disconnect-reconnect data page be returned.
Set it to $3F to request both pages.

Parameter list length
This field contains the expected length, in bytes, of the data to be returned to the host computer. A transfer length of 0 indicates that no return data is to be passed. (This is not an error condition). The scanner stops transferring data when it has returned Parameter list length bytes or all appropriate MODE SENSE data, whichever comes first.

Apple-Specific Data Page Description for the Apple Scanner

Figure 7-18 shows the Apple-specific data page for the Apple Scanner.
Figure 7-18 The MODE SENSE Apple-specific data page for the Apple Scanner

Bit Byte	7	6	5	4	3	2	1	0
0	($0B) MODE SENSE data length							
1	($00) Reserved							
2	($00) Reserved							
3	($00) Reserved							
4	($00) Page code							
5	($06) Page length							
6	($00) Reserved						Graymap	
7	Auto background-adjustment threshold							
8	($00) Reserved							Lamp
9	($00) Reserved							
10	($00) Reserved							
11	($00) Reserved							

Here is a list of the fields and bits used in this page:

MODE SENSE data length
: This field indicates the number of MODE SENSE data bytes that follow for this page. The value is $0B.

Reserved
: These fields are reserved for future expansion. They are set to 0.

Page code
: This field identifies the page type. It is set to $00.

Page length
: This field contains the length, in bytes, of the data portion of the parameter page, not including bytes 0–5. This field is set to 6.

Graymap
: This field contains the value that controls the graymap function of the scanner. The graymap function allows the scanner to enhance detail in light areas or dark areas of a document, or not to enhance either. A value of 0 results in more detail in the darker areas of the document. A value of 1 results in no alteration of the data. A value of 2 results in more detail in the lighter areas of the document.

Auto background-adjustment threshold
: This field indicates the threshold level at which the scanner determines whether a dot is black or white when the user selects automatic back- ground adjustment. The default threshold level is 64.
: When excessively dark areas of an original document are scanned, automatic background adjustment constantly adjusts the brightness level, resulting in more detail in the darker areas. Light areas are left unaltered. To select automatic background adjustment, set the Threshold parameter to 0 in the DEFINE WINDOW PARAMETERS command parameter list. See “DEFINE WINDOW PARAMETERS ($24)” later in this chapter.

Lamp
: This bit indicates the current state of the fluorescent lamp used to illuminate the document during scanning. A value of 1 indicates that the lamp is on; a value of 0 indicates that the lamp is off.

Apple-Specific Data Page Description for the OneScanner

Figure 7-19 shows the Apple-specific data page for the OneScanner.

Figure 7-19 The MODE SENSE Apple-specific data page for the OneScanner

Bit Byte	7	6	5	4	3	2	1	0
0	($0B) MODE SENSE data length							
1 - 3	($00) Reserved							
4	($00) Page code							
5	($06) Page length							
6	($00) Reserved							
7	($00) Reserved					Scan LED	Reset	Button
8	($00) Reserved				Fast H	Fast L	CCD	Lamp
9	($00) Reserved							
10	($00) Reserved							
11	($00) Reserved							

Here is a list of the fields and bits used in this page:

MODE SENSE data length
: This field indicates the number of MODE SENSE data bytes that follow for this page. The value is $0B.

Reserved
: These fields are reserved for future expansion. They are set to 0.

Page code
: This field identifies the page type. It is set to $00.

Page length
: This field contains the length, in bytes, of the data portion of the parameter page, not including bytes 0–5. This field is set to 6.

Scan LED
: This bit indicates the setting of the amber LED on the OneScanner. A value of 1 indicates that the LED is on. A value of 0 indicates that the LED is off.

Reset
: This bit indicates whether the scanner should reset the Button bit. It is used only for the MODE SELECT command. Do not use this bit. Applies only to scanners with ROM version 2.03, or earlier.

Button
: This bit indicates the current state of the button on the OneScanner since the last reset request (a MODE SELECT command with the Reset bit set to 1). A value of 1 indicates that the button has been pressed. A value of 0 indicates that the button has not been pressed. Applies only to scanners with ROM version 2.03, or earlier.

Fast H, Fast L
: Together, these bits indicate the scan speed for the scanner. Here are the three possible values for these bits:

Fast H	Fast L	Scan speed
0	0	Normal speed
0	1	High speed
1	0	Fast speed
1	1	Invalid value

At normal speed, the scanner processes the image at the slowest possible carriage speed and performs handshaking on all data exchanges with the host computer. At high speed, the scanner runs the carriage at high speed but still performs handshaking on all data exchanges with the host computer. At normal and high speeds, data handshaking prevents the loss of image data.

At fast speed, the scanner runs the carriage at high speed but does not perform any handshaking during data exchanges with the host. Eliminating the handshaking greatly improves the data exchange rate between the scanner and the host computer. However, image data may be lost if the host computer cannot keep up with the scanner. If the scanner loses data due to an overrun condition, it aborts the scan with a CHECK CONDITION status.

CCD
: This bit controls power to the CCD array in the scanner. A value of 1 causes the scanner to power on the CCD array whenever the scanner lamp is turned on. A value of 0 causes the scanner to power on the CCD array only in response to SCAN commands.

Lamp
: This bit indicates the current state of the fluorescent lamp used to illuminate the document during scanning. A value of 1 indicates that the lamp is on; a value of 0 indicates that the lamp is off.

Apple-Specific Data Page Description for the Color OneScanner

Figure 7-20 shows the Apple-specific data page for the Color OneScanner.
Figure 7-20 The MODE SENSE Apple-specific data page for the Color OneScanner

Bit / Byte	7	6	5	4	3	2	1	0
0	($0B) MODE SENSE data length							
1 - 3	($00) Reserved							
4	($00) Page code							
5	($06) Page length							
6	($00) Reserved							
7	($00) Reserved					Scan LED	($00)	($00)
8	($00) Reserved			MTF circuit	ICP bypass	Data polarity	CCD on	Reset lamp on
9	Code sense select							
10	($00) Reserved							
11	($00) Reserved							

Here is a list of the fields and bits used in this page:

MODE SENSE data length
: This field indicates the number of MODE SENSE data bytes that follow for this page. The value is $0B.

Reserved
: These fields are reserved for future expansion. They are set to 0.

Page code
: This field identifies the page type. It is set to $00.

Page length
: This field contains the length, in bytes, of the data portion of the parameter page, not including bytes 0–5. This field is set to 6.

Scan LED
: This bit indicates the setting of the amber LED which indicates the status of certain application functions. The default state for the LED is off.

MTF circuit
: This bit indicates the setting of the MTF circuit. If it is set to 1, the circuit is turned on. When it is reset to 0, the circuit is turned off.

ICP bypass
: This bit indicates the setting of the ICP. When it is set to 1, the circuit is bypassed. When it is reset to 0, the circuit is used.

Data polarity
: This bit indicates the polarity of the output image data. When it is set (1), 00 = white, and FF = black. When it is reset (0), 00 = black, and FF = white. The latter is the normal mode.

CCD on
: This bit indicates if the scanner has turned on power to the CCD (charge-coupled device) array when the scanning lamp is on. Whenever the scan command is issued, the CCD is powered up, even if the CCD on bit is set to 0.

Lamp on
: This bit, when set, indicates the scanner has turned on the lamp. The scanner goes through a shading correction before each scan, regardless of the state of the Lamp on bit. The lamp remains on until the next MODE SELECT signal is received, and the field is cleared. At this time, the lamp resumes normal on/off operation. If the lamp is on and there is no SCSI activity for more two minutes or more, the lamp turns off. When the scanner receives any SCSI command, the lamp turns on for a minimum of two minutes.

Color sense select
: This field indicates which color sensor line or lines are selected for scanning. The green line sensor is the default for grayscale scanning.

SCAN ($1B)

The SCAN command instructs the scanner to begin the scanning operation. Following the SCAN command structure, you may choose to send a Window identifier byte that identifies the scan area to process. See "DEFINE WINDOW PARAMETERS ($24)," later in this chapter, for more information.
For the Color OneScanner, there are two ways to abort a scan. The first way is to send the ABORT message. The second way is to issue the SCAN command, with the Transfer length parameter set to 0. For Apple Scanners and OneScanners, sending any command will abort the scan.

SCAN Command Structure for the Apple Scanner

Figure 7-21 shows the format of the SCAN command structure for the Apple Scanner.
Figure 7-21 The SCAN command structure for the Apple Scanner

Bit Byte	7	6	5	4	3	2	1	0
0	($1B) Operation code							
1	Logical unit number			($00) Reserved				
2	($00) Reserved							
3	($00) Reserved							
4	Transfer length							
5	Wait	No home	($00) Reserved					

Here is a list of the fields and bits used in this command structure:

Operation code
: The operation code for the SCAN command is $1B.

Reserved
: These fields are reserved for future expansion. Set them to 0.

Transfer length
: This field indicates whether the application supplies a Window identifier byte after the 6 command bytes. If the value of the transfer length is 0, the application supplies no Window identifier byte and no scan takes place. If the value of the transfer length is 1, the application supplies a Window identifier byte immediately after the command structure.
The Window identifier byte must correspond to a Window identifier field value supplied to the DEFINE WINDOW PARAMETERS command.

Wait
: If this bit is set to 1, the user must press the button on the scanner before the scanner will begin scanning.

No home
: If this bit is set to 1, the scanner lamp stays on and the carriage assembly remains where it stops at the end of a scan. After two minutes, if the scanner does not receive another SCAN command, the lamp goes off and the carriage assembly returns to the home position.

SCAN Command Structure for the OneScanner

Figure 7-22 shows the SCAN command structure for the OneScanner.

Figure 7-22 The SCAN command structure for the OneScanner

Bit Byte	7	6	5	4	3	2	1	0
0	($1B) Operation code							
1	Logical unit number			($00) Reserved				
2	($00) Reserved							
3	($00) Reserved							
4	Transfer length							
5	($00) Reserved		Non-Cal	($00) Reserved				

Here is a list of the fields and bits used in this command structure:

Operation code
: The operation code for the SCAN command is $1B.

Reserved
: These fields are reserved for future expansion. Set them to 0.

Transfer length
: This field indicates whether the application supplies a Window identifier byte after the 6 command bytes. If the value of the transfer length is 0, the application supplies no Window identifier byte and no scan takes place. If the value of the transfer length is 1, the application supplies a Window identifier byte immediately after the command structure.
: The Window identifier byte must correspond to a Window identifier field value supplied to the DEFINE WINDOW PARAMETERS command.

Non-Cal
: Controls whether the scanner calibrates for the current lamp intensity before starting the scan. If this bit is set to 1, the scanner does not perform calibration. If this bit is set to 0, the scanner calibrates for the current lamp intensity before scanning. This calibration step takes a few seconds. Your application can reduce scan time by skipping this calibration step. However, skipping lamp calibration compromises scan quality. Consequently, your application does not use calibration except for preview scans.

SCAN Command Structure for the Color OneScanner series

Figure 7-23 shows the format of the SCAN command structure for the Color OneScanner.
Figure 7-23 The SCAN command structure for the Color OneScanner

Bit Byte	7	6	5	4	3	2	1	0
0	($1B) Operation code							
1	Logical unit number			($00) Reserved				
2	($00) Reserved							
3	($00) Reserved							
4	Transfer length							
5	Control							

Here is a list of the fields and bits used in this command structure:

Operation code
: The operation code for the SCAN command is $1B.

Reserved
: These fields are reserved for future expansion. Set them to 0.

Transfer length
: This field indicates whether the application supplies a Window identifier byte after the 6 command bytes. If the value of the transfer length is 0, the application supplies no Window identifier byte and no scan takes place. If the value of the transfer length is 1, the application supplies a Window identifier byte immediately after the command structure.
The Window identifier byte must correspond to a Window identifier field value supplied to the DEFINE WINDOW PARAMETERS command.

Window Parameter List

To cause the scanner to scan a document, you must append a Window identifier byte to the SCAN command structure. The Window identifier byte specifies the scan area and related scanning parameters, and it must correspond to a window identifier defined with the DEFINE WINDOW PARAMETERS command. Figure 7-24 shows the command byte format.

Figure 7-24 The SCAN command Window identifier byte

Bit Byte	7	6	5	4	3	2	1	0
0	Window identifier							

FIELD DESCRIPTION

Window identifier
: Specifies the scan area and related parameters for the scan operation. The value of the identifier byte must be the same as the value of the Window identifier byte indicated in the DEFINE WINDOW PARAMETERS command for that window.

SEND DIAGNOSTIC ($1D)

The SEND DIAGNOSTIC command directs the scanner to run a built-in self-test. Figure 7-25 shows the format of the SEND DIAGNOSTIC command structure.

Figure 7-25 The SEND DIAGNOSTIC command structure

Bit Byte	7	6	5	4	3	2	1	0
0	($1D) Operation code							
1	Logical unit number			($00) PF	($00) Reserved	SelfTest	($00) DevOfL	($00) UnitOfL
2	($00) Reserved							
3	Parameter list (MSB)							
4	Parameter list (LSB)							
5	Control							

FIELD DESCRIPTIONS
Here is a list of the fields and bits used in this command structure:

Operation code
: The operation code for the SEND DIAGNOSTIC command is $1D.

Reserved
: These fields are reserved for future expansion. Set them to 0.

Page format (PF)
: A page format (PF) bit of one specifies that the SEND DIAGNOSTIC parameters conform to the page structure as specified in this International Standard. The implementation of the PF bit is optional. See 87.3.1 for the definition of diagnostic pages. A PF bit of zero indicates that the SEND DIAGNOSTIC parameters are as specified in SCSI-1 (i.e. all parameters are vendor-specific).

SelfTest
: Set this bit to 1 to request a self-test.

DevOfL
: A DevOfl bit of one grants permission to the target to perform diagnostic operations that may affect all the logical units on a target, e.g. alteration of reservations, log parameters, or sense data. The implementation of the DevOfl bit is optional. A DevOfL bit of zero prohibits diagnostic operations that may be detected by subsequent I/O processes.

UnitOfL
: A UnitOfL bit of one grants permission to the target to perform diagnostic operations that may affect the user accessible medium on the logical unit, e.g. write operations to the user accessible medium, or repositioning of the medium on sequential access devices. The implementation of the UnitOfl bit is optional. A UnitOfL bit of zero prohibits any diagnostic operations that may be detected by subsequent I/O processes.

The device off-line (DevOfL) and unit off-line (UnitOfL) bits are generally set by operating system software, while the parameter list is prepared by diagnostic application software. These bits grant permission to perform vendor-specific diagnostic operations on the target that may be visible to attached initiators. Thus, by preventing operations that are not enabled by these bits, the target assists the operating system in protecting its resources.

DEFINE WINDOW PARAMETERS ($24)

Before the scanner can perform a scan, the application program must provide certain details about the scan area. This information is provided in the form of parameters defining a window for each scan area. The DEFINE WINDOW PARAMETERS command passes this window-definition data to the scanner. The program defines the size, position, scanning resolution, scanning composition, and other parameters of each window. There are different parameter structures for each supported scanner. These structures are described later in this section.

IMPORTANT
The OneScanner and Color OneScanner do not support multiple windows. ▲

DEFINE WINDOW PARAMETERS Command Structure for the Apple Scanner

Figure 7-26 shows the format of the DEFINE WINDOW PARAMETERS command structure for the Apple Scanner.
Figure 7-26 The DEFINE WINDOW PARAMETERS command structure for the Apple Scanner

Bit Byte	7	6	5	4	3	2	1	0
0	($24) Operation code							
1	Logical unit number			($00) Reserved				
2 - 5	($00) Reserved							
6	Transfer length (MSB)							
7	Transfer length							
8	Transfer length (LSB)							
9	Apple	Control						

FIELD DESCRIPTIONS
Here is a list of the fields and bits used in this command structure:

Operation code
The operation code for the DEFINE WINDOW PARAMETERS command is $24.

Reserved
These fields are reserved for future expansion. Set them to 0.

Transfer length
This field contains the length, in bytes, of the DEFINE WINDOW PARAMETERS parameter list. A transfer length of 0 indicates that the application program does not pass any parameters (this is not an error condition). The transfer length indicates the total length of all window descriptors and window parameter lists. The total transfer length is

8 + (40 * n) bytes

where n is the number of windows. Byte 6 of the structure contains the most-significant byte of the transfer length; byte 8 contains the least-significant byte of the transfer length.

Apple
This bit, if set to 1, allows the Apple Scanner to recognize multiple window descriptors.

DEFINE WINDOW PARAMETERS Command Structure for the OneScanner and the Color OneScanner series

Figure 7-27 shows the format of the DEFINE WINDOW PARAMETERS command structure for the OneScanner and the Color OneScanner.

Figure 7-27 The DEFINE WINDOW PARAMETERS command structure for the OneScanner and the Color OneScanner

Bit / Byte	7	6	5	4	3	2	1	0
0	($24) Operation code							
1	Logical unit number			($00) Reserved				
2 - 5	($00) Reserved							
6	Transfer length (MSB)							
7	Transfer length							
8	Transfer length (LSB)							
9	Control							

FIELD DESCRIPTIONS

Here is a list of the fields and bits used in this command structure:

Operation code
: The operation code for the DEFINE WINDOW PARAMETERS command is $24.

Reserved
: These fields are reserved for future expansion. Set them to 0.

Transfer length
: This field contains the length, in bytes, of the DEFINE WINDOW PARAMETERS parameter list. If the value of the transfer length is 0, the application does not pass any parameters (this is not an error condition). The transfer length indicates the total length of all window descriptors and window parameter lists. The total transfer length for each model is described below. Byte 6 of the structure contains the most-significant byte of the transfer length; byte 8 contains the least-significant byte of the transfer length.

Total Transfer Length

Model	Total Transfer Length (in bytes)
Apple Scanner	48
OneScanner	48
Color OneScanner	50
Color OneScanner 600/27	72
Color OneScanner 1200/30	72

DEFINE WINDOW PARAMETERS Parameter List for the Apple Scanner

The window parameter list for the DEFINE WINDOW PARAMETERS command consists of one parameter list header and one or more unique window descriptors. When the user wants to scan more than one window, the application program must supply one window descriptor for each window. Figure 7-28 shows the format of the parameter list header and descriptor.

Figure 7-28 shows the DEFINE WINDOW PARAMETERS parameter list for the Apple Scanner

Bit / Byte	7	6	5	4	3	2	1	0
0	Data transfer length (MSB)							
1	Data transfer length (LSB)							
2	($00) Reserved							
3	($00) Reserved							
4	($00) Reserved							
5	($00) Reserved							
6	Window descriptor length (MSB)							
7	Window descriptor length (LSB)							

Bit / Byte	7	6	5	4	3	2	1	0
0	Window identifier							
1	($00) Reserved							
2 - 3	X-Axis resolution							
4 - 5	Y-Axis resolution							
6 - 9	X-Axis upper left							
10 - 13	Y-Axis upper left							
14 - 17	Window width							
18 - 21	Window length							
22	Brightness							
23	Threshold							
24	Contrast							
25	Image composition							
26	Bits per pixel							
27 - 28	Halftone pattern							
29	($00) Reserved					Padding type		
30 - 31	($00) Bit ordering							
32	Compression type							
33	($00) Compression argument							
34	($00) Reserved							
35	($00) Reserved							
36	($00) Reserved							
37	($00) Reserved							
38	($00) Reserved							
39	($00) Reserved							

FIELD DESCRIPTIONS

The parameter list header describes the length, in bytes, of a window descriptor. An application program may transfer a maximum of 99 window descriptors. Here are the parameter list fields used in this structure:

Reserved These fields are reserved for future expansion. Set them to 0.

Parameter list length
This field specifies the length in bytes of a single window descriptor. Always set this field to 40 ($28).

The window descriptor contains information about one window. Here are the window descriptors used in this structure:

Window identifier
This field contains a number between 0 and 255, which uniquely identifies the window defined by the descriptor. Use this unique identifier to identify each window during data transfers and status requests.

Reserved These fields are reserved for future expansion. Set them to 0.

X resolution This field specifies the resolution, in dots per inch, along the horizontal direction (x-axis). In Line Art mode and Halftone mode, the Apple Scanner supports resolutions of 300, 285, 270, 255, 240, 225, 210, 200, 195, 180, 165, 150, 135, 120, 105, 100, 90, and 75 dpi. In Grayscale mode, the Apple Scanner supports resolutions of 300, 200, 150, 100, and 75 dpi. A value of 0 in this field indicates that the scanner uses a default horizontal resolution value of 75 dpi.

Y resolution This field specifies the resolution, in dots per inch, in the vertical direction (y-axis). In Line Art mode and Halftone mode, the Apple Scanner supports resolutions of 300, 285, 270, 255, 240, 225, 210, 200, 195, 180, 165, 150, 135, 120, 100, 90, and 75 dpi. In Grayscale mode, the Apple Scanner supports resolutions of 300, 200, 150, 100, and 75 dpi. A value of 0 in this field indicates that the scanner uses a default vertical resolution value of 75 dpi.

Upper-left x This field specifies the x-coordinate of the upper left corner of the rectangular window to be scanned. The point (0,0) is considered the upper-left corner of the window. Coordinates are measured from this point in increments of 1/1200 of an inch from the right edge of the scanner's glass surface (as viewed when you face the glass). Coordinates must be multiples of 8 times the x resolution; they must lie on byte boundaries. The default value of this field is 0.

Upper-left y This field specifies the y-coordinate of the upper-left corner of the rectangular window to be scanned. The point (0,0) is considered the upper-left corner of the window. Coordinates are measured from this point in increments of 1/1200 of an inch from the top edge of the scanner's glass surface. The default value of this field is 0.

Width This field specifies the window width in increments of 1/1200 of an inch along the horizontal axis. This value must be a multiple of 8 times the x resolution; the window border must lie on a byte boundary. The default value of this field is 10,208.

Length This field specifies the window length in increments of 1/1200 of an inch along the vertical axis. The default value of this field is 13,200.

Brightness This field specifies the brightness. A value of 0 results in the default value of 128. Any other value indicates a relative brightness: 255 is the highest setting, 1 is the lowest setting, and 128 is the average setting.

Threshold This field specifies the threshold level. A value of 0 in Line Art mode causes the scanner to use automatic background adjustment. See "MODE SELECT ($15)," earlier in this chapter, for a description of the Auto background-adjustment threshold field. A value of 0 received in any other mode results in the use of the default setting. Any other value indicates a relative threshold parameter: 255 is the highest setting, 1 is the lowest setting, and 128 is the average setting. The default value for this parameter is 128.

Contrast This field specifies the contrast at which the scanner scans the document. A value of 0 indicates that the scanner uses the default value of 1. Any other value indicates a relative contrast: 255 is the highest setting, 1 is the lowest setting, and 128 is the average setting. The contrast setting is valid only in Halftone mode and Grayscale mode.

Image composition

This field specifies the type of image data acquired. The default composition mode is Line Art mode. The following values are valid:

$00 Line Art mode
$01 Halftone mode
$02 Grayscale mode

Bits per pixel

This field specifies, for any one pixel, the number of bits used to specify the visible density of the document being scanned. This field is valid only if the Image composition field indicates that gray-scale data is desired. The Apple Scanner supports 16 levels of gray in Grayscale mode and only black and white in Line Art mode and Halftone mode. Therefore, 4 bits per pixel are required to represent the full range of grays, and only 1 bit per pixel is required to represent either black or white.

Halftone pattern

This field specifies the halftone process by which the scanner converts multilevel gray tones to black and white dots. This parameter is only valid when the Image composition field specifies that halftone image data is desired. The following values are valid:

$00 Spiral
$01 Bayer
$02 Downloaded pattern

Padding type This field specifies what the scanner does if the image data transmitted to the host is not a whole number of bytes. This field must be $03, which truncates the data to the next byte boundary.

Bit ordering This field is resersed. Set the value to $0000.

Compression type

This field specifies the compression technique that the scanner applies to the image data prior to transmission to the host computer. The default is no compression. The following values are valid:

$00 No compression
$01 CCITT Group III, one-dimensional
$83 White line skipped

Compression argument

This field is reserved. Set the value to 0.

DEFINE WINDOW PARAMETERS Parameter List for the OneScanner

The window parameter list for the DEFINE WINDOW PARAMETERS command consists of one parameter list header followed by at most one window descriptor. Figure 7-29 shows the format of the parameter list header and descriptor for the OneScanner.

Figure 7-29 shows the DEFINE WINDOW PARAMETERS parameter list for the OneScanner

Bit / Byte	7	6	5	4	3	2	1	0
0	Data transfer length (MSB)							
1	Data transfer length (LSB)							
2	($00) Reserved							
3	($00) Reserved							
4	($00) Reserved							
5	($00) Reserved							
6	Window descriptor length (MSB)							
7	Window descriptor length (LSB)							

Bit / Byte	7	6	5	4	3	2	1	0
0	Window identifier							
1	($00) Reserved							
2 - 3	X-Axis resolution							
4 - 5	Y-Axis resolution							
6 - 9	X-Axis upper left							
10 - 13	Y-Axis upper left							
14 - 17	Window width							
18 - 21	Window length							
22	Brightness							
23	Threshold							
24	Contrast							
25	Image composition							
26	Bits per pixel							
27 - 28	Halftone pattern							
29	($00) Reserved					Padding type		
30 - 31	Scan direction							
32	Compression type							
33	Compression argument							
34	($00) Reserved							
35	($00) Reserved							
36	($00) Reserved							
37	($00) Reserved							
38	($00) Reserved							
39	($00) Reserved							

FIELD DESCRIPTIONS

The parameter list header describes the length, in bytes, of a window descriptor. An application program may transfer a maximum of one window descriptor. Here are the parameter list fields used in this structure:

Reserved — These fields are reserved for future expansion. Set them to 0.

Parameter list length
: This field specifies the length in bytes of a single window descriptor. Always set this field to 40 ($28).

The window descriptor contains information about one window. Here are the window descriptors used in this structure:

Window identifier
: This field identifies the window defined by the descriptor. This field must be set to 0.

Reserved — These fields are reserved for future expansion. Set them to 0.

X resolution — This field specifies the resolution, in dots per inch, along the horizontal direction (x-axis). The OneScanner supports horizontal resolutions that range from 72 to 300 dpi in 1-dpi increments. A value of 0 in this field indicates that the scanner uses a default horizontal resolution value of 72 dpi.

Y resolution — This field specifies the resolution, in dots per inch, in the vertical direction (y-axis). The OneScanner supports vertical resolutions that range from 72 to 300 dpi in 1-dpi increments. A value of 0 in this field indicates that the scanner uses a default vertical resolution value of 72 dpi.

Upper-left x This field specifies the x-coordinate of the upper left corner of the rectangular window to be scanned. The point (0,0) is considered the upper-left corner of the window. Coordinates are measured from this point in increments of 1/1200 of an inch from the right edge of the scanner's glass surface (as viewed when you face the glass). Coordinates must be multiples of 8 times the x resolution; they must lie on byte boundaries. The default value of this field is 0.

Upper-left y This field specifies the y-coordinate of the upper-left corner of the rectangular window to be scanned. The point (0,0) is considered the upper-left corner of the window. Coordinates are measured from this point in increments of 1/1200 of an inch from the top edge of the scanner's glass surface. The default value of this field is 0.

Width This field specifies the window width in increments of 1/1200 of an inch along the horizontal axis. This value must be a multiple of 8 times the x resolution; the window border must lie on a byte boundary. The default value of this field is 10,200.

Length This field specifies the window length in increments of 1/1200 of an inch along the vertical axis. The default value of this field is 13,200.

Brightness This field specifies the brightness. A value of 0 results in the default value of 128. Any other value indicates a relative brightness: 255 is the highest setting, 1 is the lowest setting, and 128 is the average setting.

Threshold This field specifies the threshold level. A value of 0 in Line Art mode causes the scanner to use automatic background adjustment. See "MODE SELECT ($15)," earlier in this chapter, for a description of the Auto background-adjustment threshold field. A value of 0 received in any other mode results in the use of the default setting. Any other value indicates a relative threshold parameter: 255 is the highest setting, 1 is the lowest setting, and 128 is the average setting. The default value for this parameter is 128.

Contrast This field specifies the contrast at which the scanner scans the document. A value of 0 indicates that the scanner uses the default value of 1. Any other value indicates a relative contrast: 255 is the highest setting, 1 is the lowest setting, and 128 is the average setting. The contrast setting is valid only in Halftone mode and Grayscale mode.

Image composition

This field specifies the type of image data acquired. The default composition mode is Line Art mode. The following values are valid:

$00 Line Art mode
$01 Halftone mode
$02 Grayscale mode

Bits per pixel

This field specifies, for any one pixel, the number of bits used to specify the visible density of the document being scanned. This field is valid only if the Image composition field indicates that gray-scale data is desired. The OneScanner supports either 16 or 256 levels of gray in Grayscale mode. For 16-level gray support, set this field to 4; for 256-level gray support, set this field to 8.

Halftone pattern

This field specifies the halftone process by which the scanner converts multilevel gray tones to black and white dots. This parameter is only valid when the Image composition field specifies that halftone image data is desired. The following values are valid:

$00 4-by-4 Spiral
$01 4-by-4 Bayer
$02 Downloaded pattern
$03 8-by-8 Spiral
$04 8-by-8 Bayer

Padding type This field specifies what the scanner does if the image data transmitted to the host is not a whole number of bytes. This field must be $03, which truncates the data to the next byte boundary.

Scan direction

This field specifies the direction for the scan operation. This field must be set to 0, which specifies that the scan is to proceed from left to right and top to bottom.

Compression type

This field specifies the compression technique that the scanner applies to the image data prior to transmission to the host computer. This field must be set to 0, which indicates that the scanner is not to compress the image data.

Compression argument

This field is reserved. Set the value to 0.

DEFINE WINDOW PARAMETERS Parameter List for the Color OneScanner

The window parameter list for the DEFINE WINDOW PARAMETERS command consists of one parameter list header followed by at most one window descriptor. Figure 7-30 shows the format of the parameter list header and descriptor for the Color OneScanner.

Figure 7-30 shows the DEFINE WINDOW PARAMETERS parameter list for the Color OneScanner

Bit Byte	7	6	5	4	3	2	1	0
0	Data transfer length (MSB)							
1	Data transfer length (LSB)							
2	($00) Reserved							
3	($00) Reserved							
4	($00) Reserved							
5	($00) Reserved							
6	Window descriptor length (MSB)							
7	Window descriptor length (LSB)							

Bit Byte	7	6	5	4	3	2	1	0
0	Window identifier							
1	($00) Reserved							
2 - 3	X-Axis resolution							
4 - 5	Y-Axis resolution							
6 - 9	X-Axis upper left							
10 - 13	Y-Axis upper left							
14 - 17	Window width							
18 - 21	Window length							
22	Brightness							
23	Threshold							
24	Contrast							
25	Image composition							
26	Bits per pixel							
27 - 28	($00) Halftone pattern							
29	($00) Reserved					Padding type		
30 - 31	($00) Bit ordering							
32	Compression type							
33	Compression argument							
34	($00) Reserved							
35	($00) Reserved							
36	($00) Reserved							
37	($00) Reserved							
38	($00) Reserved							
39	($00) Reserved							
40	Digital data for VRT							
41	Digital data for VRB							

FIELD DESCRIPTIONS

The parameter list header describes the length, in bytes, of a window descriptor. An application program may transfer a maximum of one window descriptor. Here are the parameter list fields used in this structure:

Reserved These fields are reserved for future expansion. Set them to 0.

Parameter list length
This field specifies the length in bytes of a single window descriptor. Always set this field to 42 ($2A).

The window descriptor contains information about one window. Here are the window descriptors used in this structure:

Window identifier
This field identifies the window defined by the descriptor. This field must be set to 0.

Reserved These fields are reserved for future expansion. Set them to 0.

X resolution This field specifies the resolution, in dots per inch, along the horizontal direction (x-axis). The Color OneScanner supports horizontal resolutions that range from 72 to 300 dpi in 1-dpi increments. A value of 0 in this field indicates that the scanner uses a default horizontal resolution value of 72 dpi.

Y resolution This field specifies the resolution, in dots per inch, in the vertical direction (y-axis). The Color OneScanner supports vertical resolutions that range from 72 to 300 dpi in 1-dpi increments. A value of 0 in this field indicates that the scanner uses a default vertical resolution value of 72 dpi.

Upper-left x This field specifies the x-coordinate of the upper left corner of the rectangular window to be scanned. The point (0,0) is considered the upper-left corner of the window. Coordinates are measured from this point in increments of 1/1200 of an inch from the right edge of the scanner's glass surface (as viewed when you face the glass). Coordinates must be multiples of 8 times the x resolution; they must lie on byte boundaries. The default value of this field is 0.

Upper-left y This field specifies the y-coordinate of the upper-left corner of the rectangular window to be scanned. The point (0,0) is considered the upper-left corner of the window. Coordinates are measured from this point in increments of 1/1200 of an inch from the top edge of the scanner's glass surface. The default value of this field is 0.

Width This field specifies the window width in increments of 1/1200 of an inch along the horizontal axis. This value must be a multiple of 8 times the x resolution; the window border must lie on a byte boundary. The default value of this field is 10,200.

Length This field specifies the window length in increments of 1/1200 of an inch along the vertical axis. The default value of this field is 13,200.

Brightness This field specifies the brightness. A value of 0 results in the default value of 128. Any other value indicates a relative brightness: 255 is the highest setting, 1 is the lowest setting, and 128 is the average setting.

Threshold This field specifies the threshold level. A value of 0 in Line Art mode causes the scanner to use automatic background adjustment. See "MODE SELECT ($15)," earlier in this chapter, for a description of the Auto background-adjustment threshold field. A value of 0 received in any other mode results in the use of the default setting. Any other value indicates a relative threshold parameter: 255 is the highest setting, 1 is the lowest setting, and 128 is the average setting. The default value for this parameter is 128.

Contrast This field specifies the contrast at which the scanner scans the document. A value of 0 indicates that the scanner uses the default value of 1. Any other value indicates a relative contrast: 255 is the highest setting, 1 is the lowest setting, and 128 is the average setting. The contrast setting is valid only in Halftone mode and Grayscale mode.

Image composition

This field specifies the type of image data acquired. The default composition mode is Line Art mode. The following values are valid:

$00 Line Art mode (bi-level black and white)
$02 Grayscale mode (multi-level black and white)
$03 Bi-level Color mode (RGB)
$05 Full Color mode (RGB)

Bits per pixel

This field specifies, for any one pixel, the number of bits used to specify the visible density of the document being scanned. This field is only valid if the Image composition field indicates grayscale data is desired. The Color OneScanner supports 4 ($04) and 8 ($08) bits of gray-scale or color data, 3 bits for bi-level color, and 24 bits for full color (8 bits per color). For 16-level gray support, set this field to 4; for 256-level gray support, set this field to 8.

Halftone pattern

This field is reserved. Set the value to $0000.

Padding type This field specifies what the scanner does if the image data transmitted to the host is not a whole number of bytes. This field must be $03, which truncates the data to the next byte boundary.

Bit ordering This field is reserved. Set the value to $0000.

Compression type

This field specifies the compression technique that the scanner applies to the image data prior to transmission to the host computer. This field must be set to 0, which indicates that the scanner is not to compress the image data.

Compression argument

This field is reserved. Set the value to 0.

Digital data for Vrt and Vrb

These fields determine the contrast range of a scanned image. Vrt (voltage reference top) determines the upper end of the contrast range, and Vrb (voltage reference bottom) the lower end of the contrast range. If there is a value of 0 in either Vrt or Vrb, Vrt and Vrb are disabled, and Brightness and Contrast are used instead. Any other setting between 1 and 255 indicates the scanner should use the related value, for example, if the setting is 3, the scanner should use the value 3, and so forth.

DEFINE WINDOW PARAMETERS Parameter List for the Color OneScanner 600/27

The window parameter list for the DEFINE WINDOW PARAMETERS command consists of one parameter list header followed by at most one window descriptor. Figure 7-30B shows the format of the parameter list header and descriptor for the Color OneScanner 600/27.

Figure 7-30B shows the DEFINE WINDOW PARAMETERS parameter list for the Color OneScanner 600/27

Bit Byte	7	6	5	4	3	2	1	0
0	Data transfer length (MSB)							
1	Data transfer length (LSB)							
2	($00) Reserved							
3	($00) Reserved							
4	($00) Reserved							
5	($00) Reserved							
6	Window descriptor length (MSB)							
7	Window descriptor length (LSB)							

Bit Byte	7	6	5	4	3	2	1	0
0	Window identifier							
1	($00) Reserved							
2 - 3	X-Axis resolution							
4 - 5	Y-Axis resolution							
6 - 9	X-Axis upper left							
10 - 13	Y-Axis upper left							
14 - 17	Window width							
18 - 21	Window length							
22	Brightness							
23	Threshold							
24	Contrast							
25	Image composition							
26	Bits per pixel							
27 - 28	($00) Halftone pattern							
29	RIF	($00) Reserved				Padding type		
30 - 31	($00) Bit ordering							
32	Compression type							
33	Compression argument							
34	($00) Reserved							
35	($00) Reserved							
36	($00) Reserved							
37	($00) Reserved							
38	($00) Reserved							
39	($00) Reserved							
40	Digital data for VRT							
41	Digital data for VRB							
42	Canon settings							
43	($00) Reserved							
44	($00) Reserved							
45	($00) Reserved							
46	($02) Reserved							
47	($00) Reserved							
48	($00) Reserved							
49	($01) Reserved							
50	($01) Reserved							
51	Hilite level for Red color							
52	Shadow level for Red color							
53	($00) Reserved							
54	Hilite level for Green color							
55	($00) Reserved							
56	Shadow level for Green colur							
57	($00) Reserved							
58	($00) Reserved							
59	($00) Reserved							
60	($00) Reserved							
61	($00) Reserved							
62	Hilite level for Blue color							
63	Shadow level for Blue color							

FIELD DESCRIPTIONS

The parameter list header describes the length, in bytes, of a window descriptor. An application program may transfer a maximum of one window descriptor. Here are the parameter list fields used in this structure:

Reserved — These fields are reserved for future expansion. Set them to 0.

Parameter list length — This field specifies the length in bytes of a single window descriptor. Always set this field to 64 ($40).

The window descriptor contains information about one window. Here are the window descriptors used in this structure:

Window identifier — This field identifies the window defined by the descriptor. This field must be set to 0.

Reserved — These fields are reserved for future expansion. Set them to 0.

X resolution — This field specifies the resolution, in dots per inch, along the horizontal direction (x-axis). The Color OneScanner 600/27 supports horizontal resolutions that range from 72 to 300 dpi in 1-dpi increments. A value of 0 in this field indicates that the scanner uses a default horizontal resolution value of 72 dpi.

Y resolution — This field specifies the resolution, in dots per inch, in the vertical direction (y-axis). The Color OneScanner 600/27 supports vertical resolutions that range from 72 to 600 dpi in 1-dpi increments. A value of 0 in this field indicates that the scanner uses a default vertical resolution value of 72 dpi.

Upper-left x — This field specifies the x-coordinate of the upper left corner of the rectangular window to be scanned. The point (0,0) is considered the upper-left corner of the window. Coordinates are measured from this point in increments of 1/1200 of an inch from the right edge of the scanner's glass surface (as viewed when you face the glass). Coordinates must be multiples of 8 times the x resolution; they must lie on byte boundaries. The default value of this field is 0.

Upper-left y — This field specifies the y-coordinate of the upper-left corner of the rectangular window to be scanned. The point (0,0) is considered the upper-left corner of the window. Coordinates are measured from this point in increments of 1/1200 of an inch from the top edge of the scanner's glass surface. The default value of this field is 0.

Width — This field specifies the window width in increments of 1/1200 of an inch along the horizontal axis. This value must be a multiple of 8 times the x resolution; the window border must lie on a byte boundary. The default value of this field is 10,200.

Length — This field specifies the window length in increments of 1/1200 of an inch along the vertical axis. The default value of this field is 13,200.

Brightness — This field specifies the brightness. A value of 0 results in the default value of 128. Any other value indicates a relative brightness: 255 is the highest setting, 1 is the lowest setting, and 128 is the average setting.

Threshold — This field specifies the threshold level. A value of 0 in Line Art mode causes the scanner to use automatic background adjustment. See "MODE SELECT ($15)," earlier in this chapter, for a description of the Auto background-adjustment threshold field. A value of 0 received in any other mode results in the use of the default setting. Any other value indicates a relative threshold parameter: 255 is the highest setting, 1 is the lowest setting, and 128 is the average setting. The default value for this parameter is 128.

Contrast — This field specifies the contrast at which the scanner scans the document. A value of 0 indicates that the scanner uses the default value of 1. Any other value indicates a relative contrast: 255 is the highest setting, 1 is the lowest setting, and 128 is the average setting. The contrast setting is valid only in Halftone mode and Grayscale mode.

Image composition — This field specifies the type of image data acquired. The default composition mode is Line Art mode. The following values are valid:

- $00 Line Art mode (bi-level black and white)
- $02 Grayscale mode (multi-level black and white)
- $03 Bi-level Color mode (RGB)
- $05 Full Color mode (RGB)

Bits per pixel — This field specifies, for any one pixel, the number of bits used to specify the visible density of the document being scanned. This field is only valid if the Image composition field indicates grayscale data is desired. The Color OneScanner 600/27 supports 4 ($04) and 8 ($08) bits of gray-scale or color data, 3 bits for bi-level color, and 24 bits for full color (8 bits per color). For 16-level gray support, set this field to 4; for 256-level gray support, set this field to 8.

Halftone pattern — This field is reserved. Set the value to $0000.

Reverse Image Format (RIF)
: This field specifies if the source document is a negative film or a normal document. Set the bit to 0 if the source document is a negative film, or set the bit to 1 if the source document is a normal document

Padding type
: This field specifies what the scanner does if the image data transmitted to the host is not a whole number of bytes. This field must be $03, which truncates the data to the next byte boundary.

Bit ordering
: This field is reserved. Set the value to $0000.

Compression type
: This field specifies the compression technique that the scanner applies to the image data prior to transmission to the host computer. This field must be set to 0, which indicates that the scanner is not to compress the image data.

Compression argument
: This field is reserved. Set the value to 0.

Digital data for Vrt and Vrb
: These fields determine the contrast range of a scanned image. Vrt (voltage reference top) determines the upper end of the contrast range, and Vrb (voltage reference bottom) the lower end of the contrast range. If there is a value of 0 in either Vrt or Vrb, Vrt and Vrb are disabled, and Brightness and Contrast are used instead. Any other setting between 1 and 255 indicates the scanner should use the related value, for example, if the setting is 3, the scanner should use the value 3, and so forth.

Canon settings
: This field specifies some Canon-specific parameters for scanning.
Gray Response Curve, bit 3 - Set to 0 if you do not activate the gray response curve. Set to 1 if you activate the gray response curve.
Mirror, bit 2 - Set to 0 if the image is returned as it is scanned. Set to 1 if the returned image will be returned mirrored.
Auto-Exposure, bit 0 - Set to 0 if you do not acivate the auto-exposure setting. Set to 1 if you activate the auto-exposure setting.

Highlight and Shadow levels
: Highlights are the lightest elements in an image, whereas shadows are the darkest areas. You can set a highlight and a shadow value in the range from 0 to 255 for each of the three components R, G, and B.

DEFINE WINDOW PARAMETERS Parameter List for the Color OneScanner 1200/30

The window parameter list for the DEFINE WINDOW PARAMETERS command consists of one parameter list header followed by at most one window descriptor. Figure 7-30B shows the format of the parameter list header and descriptor for the Color OneScanner 1200/30.

Figure 7-30C shows the DEFINE WINDOW PARAMETERS parameter list for the Color OneScanner 1200/30

Bit Byte	7	6	5	4	3	2	1	0
0	Data transfer length (MSB)							
1	Data transfer length (LSB)							
2	($00) Reserved							
3	($00) Reserved							
4	($00) Reserved							
5	($00) Reserved							
6	Window descriptor length (MSB)							
7	Window descriptor length (LSB)							

Bit Byte	7	6	5	4	3	2	1	0
0	Window identifier							
1	($00) Reserved							
2 - 3	X-Axis resolution							
4 - 5	Y-Axis resolution							
6 - 9	X-Axis upper left							
10 - 13	Y-Axis upper left							
14 - 17	Window width							
18 - 21	Window length							
22	Brightness							
23	Threshold							
24	Contrast							
25	Image composition							
26	Bits per pixel							
27 - 28	($00) Halftone pattern							
29	RIF	($00) Reserved				Padding type		
30 - 31	($00) Bit ordering							
32	Compression type							
33	Compression argument							
34	($00) Reserved							
35	($00) Reserved							
36	($00) Reserved							
37	($00) Reserved							
38	($00) Reserved							
39	($00) Reserved							
40	Digital data for VRT							
41	Digital data for VRB							
42	Canon settings							
43	($00) Reserved							
44	($00) Reserved							
45	($00) Reserved							
46	($02) Reserved							
47	($00) Reserved							
48	($00) Reserved							
49	($01) Reserved							
50	($01) Reserved							
51	Hilite level for Red color							
52	Shadow level for Red color							
53	($00) Reserved							
54	Hilite level for Green color							
55	($00) Reserved							
56	Shadow level for Green colur							
57	($00) Reserved							
58	($00) Reserved							
59	($00) Reserved							
60	($00) Reserved							
61	($00) Reserved							
62	Hilite level for Blue color							
63	Shadow level for Blue color							

FIELD DESCRIPTIONS

The parameter list header describes the length, in bytes, of a window descriptor. An application program may transfer a maximum of one window descriptor. Here are the parameter list fields used in this structure:

Reserved These fields are reserved for future expansion. Set them to 0.

Parameter list length
This field specifies the length in bytes of a single window descriptor. Always set this field to 64 ($40).

The window descriptor contains information about one window. Here are the window descriptors used in this structure:

Window identifier
This field identifies the window defined by the descriptor. This field must be set to 0.

Reserved These fields are reserved for future expansion. Set them to 0.

X resolution This field specifies the resolution, in dots per inch, along the horizontal direction (x-axis). The Color OneScanner 1200/30 supports horizontal resolutions that range from 72 to 600 dpi in 1-dpi increments. A value of 0 in this field indicates that the scanner uses a default horizontal resolution value of 72 dpi.

Y resolution This field specifies the resolution, in dots per inch, in the vertical direction (y-axis). The Color OneScanner 1200/30 supports vertical resolutions that range from 72 to 1200 dpi in 1-dpi increments. A value of 0 in this field indicates that the scanner uses a default vertical resolution value of 72 dpi.

Upper-left x This field specifies the x-coordinate of the upper left corner of the rectangular window to be scanned. The point (0,0) is considered the upper-left corner of the window. Coordinates are measured from this point in increments of 1/1200 of an inch from the right edge of the scanner’s glass surface (as viewed when you face the glass). Coordinates must be multiples of 8 times the x resolution; they must lie on byte boundaries. The default value of this field is 0.

Upper-left y This field specifies the y-coordinate of the upper-left corner of the rectangular window to be scanned. The point (0,0) is considered the upper-left corner of the window. Coordinates are measured from this point in increments of 1/1200 of an inch from the top edge of the scanner’s glass surface. The default value of this field is 0.

Width This field specifies the window width in increments of 1/1200 of an inch along the horizontal axis. This value must be a multiple of 8 times the x resolution; the window border must lie on a byte boundary. The default value of this field is 10,200.

Length This field specifies the window length in increments of 1/1200 of an inch along the vertical axis. The default value of this field is 13,200.

Brightness This field specifies the brightness. A value of 0 results in the default value of 128. Any other value indicates a relative brightness: 255 is the highest setting, 1 is the lowest setting, and 128 is the average setting.

Threshold This field specifies the threshold level. A value of 0 in Line Art mode causes the scanner to use automatic background adjustment. See “MODE SELECT ($15),” earlier in this chapter, for a description of the Auto background-adjustment threshold field. A value of 0 received in any other mode results in the use of the default setting. Any other value indicates a relative threshold parameter: 255 is the highest setting, 1 is the lowest setting, and 128 is the average setting. The default value for this parameter is 128.

Contrast This field specifies the contrast at which the scanner scans the document. A value of 0 indicates that the scanner uses the default value of 1. Any other value indicates a relative contrast: 255 is the highest setting, 1 is the lowest setting, and 128 is the average setting. The contrast setting is valid only in Halftone mode and Grayscale mode.

Image composition
This field specifies the type of image data acquired. The default composition mode is Line Art mode. The following values are valid:
$00 Line Art mode (bi-level black and white)
$02 Grayscale mode (multi-level black and white)
$03 Bi-level Color mode (RGB)
$05 Full Color mode (RGB)

Bits per pixel
This field specifies, for any one pixel, the number of bits used to specify the visible density of the document being scanned. This field is only valid if the Image composition field indicates grayscale data is desired. The Color OneScanner 1200/30 supports 4 ($04) and 8 ($08) bits of gray-scale or color data, 3 bits for bi-level color, and 24 bits for full color (8 bits per color). For 16-level gray support, set this field to 4; for 256-level gray support, set this field to 8.

Halftone pattern
This field is reserved. Set the value to $0000.

Reverse Image Format (RIF)
This field specifies if the source document is a negative film or a normal document. Set the bit to 0 if the source document is a negative film, or set the bit to 1 if the source document is a normal document

Padding type This field specifies what the scanner does if the image data transmitted to the host is not a whole number of bytes. This field must be $03, which truncates the data to the next byte boundary.

Bit ordering This field is reserved. Set the value to $0000.

Compression type
This field specifies the compression technique that the scanner applies to the image data prior to transmission to the host computer. This field must be set to 0, which indicates that the scanner is not to compress the image data.

Compression argument
This field is reserved. Set the value to 0.

Digital data for Vrt and Vrb
These fields determine the contrast range of a scanned image. Vrt (voltage reference top) determines the upper end of the contrast range, and Vrb (voltage reference bottom) the lower end of the contrast range. If there is a value of 0 in either Vrt or Vrb, Vrt and Vrb are disabled, and Brightness and Contrast are used instead. Any other setting between 1 and 255 indicates the scanner should use the related value, for example, if the setting is 3, the scanner should use the value 3, and so forth.

Canon settings
This field specifies some Canon-specific parameters for scanning.
Gray Response Curve, bit 3 - Set to 0 if you do not activate the gray response curve. Set to 1 if you activate the gray response curve.
Mirror, bit 2 - Set to 0 if the image is returned as it is scanned. Set to 1 if the returned image will be returned mirrored.
Auto-Exposure, bit 0 - Set to 0 if you do not acivate the auto-exposure setting. Set to 1 if you activate the auto-exposure setting.

Highlight and Shadow levels
Highlights are the lightest elements in an image, whereas shadows are the darkest areas. You can set a highlight and a shadow value in the range from 0 to 255 for each of the three components R, G, and B.

Using Window Descriptors With the Apple Scanner

Note
This section applies only to the Apple Scanner.

If the Apple bit in a DEFINE WINDOW PARAMETERS command structure is set to 1, an application program may send more than one window descriptor. The first window descriptor defines the primary window as well as the parameters for the entire scan. The additional window descriptors define secondary windows within the primary window, each of which may have a different image composition type. For example, the primary window could be used to define a scan as Line Art mode, and you may also define secondary windows within the primary window as Halftone composition.

Since halftone data is basically just a black-and-white image, the halftone data and line art data can be mixed freely. However, only halftone data and line art data can be mixed. Mixing with grayscale data is not allowed. If windows overlap, the window descriptor with the higher window identifier value takes priority.

Additional window descriptors must have all fields set to the same value as those in the primary window descriptor, except for the five fields that define the window rectangle (upper-left y, upper-left x, Width, Length, and Window identifier) and the Composition field. Window descriptor fields with different values result in an error. The byte- boundary restriction that applies to the parameters Width and upper-left x of the primary window does not apply to those parameters for secondary windows.

You may specify any window identifier as a parameter in the DEFINE WINDOW PARAMETERS command. All other commands—such as SCAN, GET DATA BUFFER STATUS, and READ—accept only the window identifier of a primary window.

Structure details

Image composition codes

Code	Description
00h	Bi-level black & white
01h	Dithered/halftone black & white
02h	Multi-level black & white (gray scale)
03h	Bi-level RGB colour
04h	Dithered/halftone RGB colour
05h	Multi-level RGB colour
06h - FFh	Reserved

Padding types

Code	Description
00h	No padding
01h	Pad with 0's to byte boundary
02h	Pad with 1's to byte boundary
03h	Truncate to byte boundary
04h - FFh	Reserved

Compression types and arguments

Code	Description	Compression argument
00h	No compression	Reserved
01h	CCITT group III, 1 dimensional	Reserved
02h	CCITT group III, 2 dimensional	K factor
03h	CCITT group IV, 2 dimensional	Reserved
04h - 0Fh	Reserved	Reserved
10h	Optical character recognition (OCR)	Vendor-specific
11h - 7Fh	Reserved	Reserved
80h - FFh	Vendor-specific	Vendor-specific

GET WINDOW PARAMETERS ($25)

The GET WINDOWS PARAMETERS command provides a means for the initiator to get information about a window previously defined using the DEFINE WINDOW PARAMETERS command. Figure 7-31 shows the format of the GET WINDOW PARAMETERS command structure.

Note
This command is supported only by the Color OneScanner.

Figure 7-31 The GET WINDOW PARAMETERS command structure

Bit / Byte	7	6	5	4	3	2	1	0
0	($25) Operation code							
1	Logical unit number			($00) Reserved				Single
2	($00) Reserved							
3	($00) Reserved							
4	($00) Reserved							
5	($00) Window identifier							
6	Transfer length (MSB)							
7	Transfer length							
8	Transfer length (LSB)							
9	Control							

FIELD DESCRIPTIONS
Here is a list of the fields and bits used in this command structure:

Operation code
: The operation code for the GET WINDOW PARAMETERS command is $25.

Reserved
: These fields are reserved for future expansion. Set them to 0.

Single
: A single bit of one specifies that a single window descriptor shall be returned for the specified window identifier. A single bit of zero specifies that window descriptors be returned for all window identifiers that were defined by a SET WINDOWS command or by the target, if the automatic bit was set to one.

Window identifier
: The only valid value for this field is $00.

Transfer length
: These fields specify the length, in bytes, of the window description information sent to the initiator. If the value of the transfer length is 0, no window description information was sent. This is not considered an error. The scanner stops transferring data when it has returned data equivalent to the transfer byte length, or when it has returned all appropriate GET WINDOWS data, whichever comes first.

Figure 7-32 shows the GET WINDOW PARAMETERS parameter list.

Bit Byte	7	6	5	4	3	2	1	0
0	Data transfer length (MSB)							
1	Data transfer length (LSB)							
2	($00) Reserved							
3	($00) Reserved							
4	($00) Reserved							
5	($00) Reserved							
6	($00) Window descriptor length (MSB)							
7	($2A) Window descriptor length (LSB)							

Bit Byte	7	6	5	4	3	2	1	0
0	Window identifier							
1	($00) Reserved							
2 - 3	X-Axis resolution							
4 - 5	Y-Axis resolution							
6 - 9	X-Axis upper left							
10 - 13	Y-Axis upper left							
14 - 17	Window width							
18 - 21	Window length							
22	Brightness							
23	Threshold							
24	Contrast							
25	Image composition							
26	Bits per pixel							
27 - 28	Halftone pattern							
29	RIF	($00) Reserved				Padding type		
30 - 31	Bit ordering							
32	Compression type							
33	Compression argument							
34	($00) Reserved							
35	($00) Reserved							
36	($00) Reserved							
37	($00) Reserved							
38	($00) Reserved							
39	($00) Reserved							
40	DD (Digital data) for VRT							
41	DD (Digital data) for VRB							

Refer to the DEFINE WINDOW PARAMETERS section above for detailed information of each field.

READ ($28)

The READ command instructs the scanner to send data currently in its memory to the host computer. With the Apple Scanner, the READ command may return only image data. With the OneScanner, the READ command may return image data or the contents of the scanner's calibration RAM or static RAM. The Color OneScanner may return image data, the 3-by-3 matrix for color correction, the gamma function table, and PSRAM data.
Use the GET DATA STATUS command to determine the amount of image data available for a particular window. See "GET DATA STATUS ($34)," later in this chapter, for more information.

READ Command Structure for the Apple Scanner

Figure 7-33 shows the format of the READ command structure for the Apple Scanner.
Figure 7-33 The READ command structure for the Apple Scanner

Bit / Byte	7	6	5	4	3	2	1	0
0	($28) Operation code							
1	Logical unit number			($00) Reserved				
2	($00) Data type code							
3	($00) Reserved							
4	Data type qualifier (MSB)							
5	Data type qualifier (LSB)							
6	Transfer length (MSB)							
7	Transfer length							
8	Transfer length (LSB)							
9	Control							

FIELD DESCRIPTIONS
Here is a list of the fields used in this command structure:

Operation code
: The operation code for the READ command is $28.

Reserved
: These fields are reserved for future expansion. Set them to 0.

Data type code
: This field specifies the type of data to be read. It must be set to $00.

Data type qualifier
: This field specifies the window for an image data read operation. Before the scanner can read data from a window, your application program must define the window in a DEFINE WINDOW PARAMETERS command and then include it in the window list of a SCAN command.

Transfer length
: This field specifies the maximum number of bytes that the host computer has allocated for the returned data. If the allocation length is 0, the scanner transfers no data (this is not an error condition). The command is terminated either when the host computer receives Allocation length bytes or when the scanner sends all available data, whichever is less. The GET DATA STATUS command determines the amount of data available for a given window.

READ Command Structure for the OneScanner

Figure 7-34 shows the READ command structure for the OneScanner. When reading the OneScanner calibration RAM, you must issue a read request for 2,550 bytes. When reading a downloaded halftone pattern from the OneScanner static RAM, you may read all or part of the pattern.

Figure 7-34 The READ command structure for the OneScanner

Bit / Byte	7	6	5	4	3	2	1	0
0	($28) Operation code							
1	Logical unit number			($00) Reserved				
2	($00 or $02 or $80) Data type code							
3	($00) Reserved							
4	($00) Data type qualifier (MSB)							
5	($00) Data type qualifier (LSB)							
6	Transfer length (MSB)							
7	Transfer length							
8	Transfer length (LSB)							
9	Control							

FIELD DESCRIPTIONS

Here is a list of the fields used in this command structure:

Operation code
: The operation code for the READ command is $28.

Reserved
: These fields are reserved for future expansion. Set them to 0.

Data type code
: This field specifies the type of data to be read. Set this field to one of the following values:
$00 Read image data
$02 Read halftone pattern from static RAM
$80 Read calibration RAM

Data type qualifier
: This field specifies the window for an image data read operation. Before the scanner can read data from a window, your application program must define the window in a DEFINE WINDOW PARAMETERS command and then include it in the window list of a SCAN command.

Transfer length
: This field specifies the maximum number of bytes that the host computer has allocated for the returned data. If the allocation length is 0, the scanner transfers no data (this is not an error condition). The command is terminated either when the host computer receives Allocation length bytes or when the scanner sends all available data, whichever is less. The GET DATA STATUS command determines the amount of data available for a given window.
When reading the contents of the processor static RAM (Transfer data type set to $02), you may read some or all of a downloaded halftone pattern. Set Allocation length to a value less than or equal to the size of the stored halftone matrix plus the dimension byte (see "Halftone Parameter Page Description," later in this chapter, for a detailed description of the format of a halftone matrix). You may not read the contents of any of the built-in halftone patterns.
When reading the contents of calibration RAM (Transfer data type set to $80), you may read some or all of that memory. Set Allocation length to a value in the range from 1 through 255. Each returned byte contains the calibration data for the corresponding pixel sensor in the CCD array.

READ Command Structure for the Color OneScanner

Figure 7-35 shows the READ command structure for the Color OneScanner. When reading the Color OneScanner calibration RAM, you must issue a read request for 8,100 bytes. When reading the 3-by-3 matrix table, you must issue a read request for 18 bytes. When reading the gamma table, you must issue a read request for 768 bytes.

Figure 7-35 The READ command structure for the Color OneScanner

Bit Byte	7	6	5	4	3	2	1	0
0	($28) Operation code							
1	Logical unit number			($00) Reserved				
2	($00 or $01 or $03 or $80) Data type code							
3	($00) Reserved							
4	($00) Data type qualifier (MSB)							
5	($00 or $02) Data type qualifier (LSB)							
6	Transfer length (MSB)							
7	Transfer length							
8	Transfer length (LSB)							
9	Control							

FIELD DESCRIPTIONS

Here is a list of the fields used in this command structure:

Operation code
: The operation code for the READ command is $28.

Reserved
: These fields are reserved for future expansion. Set them to 0.

Data type code
: This field specifies the type of data to be read. Set this field to one of the following values:
$00 Read image data
$01 Read 3-by-3 matrix for color correction
$03 Read gamma function table
$80 Read calibration RAM

Data type qualifier
: This field specifies the window for an image data read operation. Before the scanner can read data from a window, your application program must define the window in a DEFINE WINDOW PARAMETERS command and then include it in the window list of a SCAN command. The first byte of the Transfer ID field is set to $00, meaning only one window is supported. The second byte of the Transfer ID field differentiates between data transfers of the same Transfer data type, as shown in Table 7-2.

Transfer length
: This field specifies the maximum number of bytes that the host computer has allocated for the returned data. If the allocation length is 0, the scanner transfers no data (this is not an error condition). The command is terminated either when the host computer receives Allocation length bytes or when the scanner sends all available data, whichever is less. The GET DATA STATUS command determines the amount of data available for a given window.
When reading the contents of the calibration RAM (Transfer data type set to $80), you may read some or all of that memory. Set Allocation length to 8100. Each returned byte contains the calibration data for the correspond- ing pixel sensor in the CCD array. When reading the 3-by-3 matrix table, set the Allocation length to 18 bytes. When reading the Gamma table, set the Allocation length to 768 bytes
When reading image data, the data transferred must always be an even number of bytes. If the initiator asks for an odd number of bytes, the scanner returns a CHECK CONDITION status. If the initiator then sends a REQUEST SENSE command, the scanner returns SENSE KEY $5, indicating that this is an illegal transaction. If the initiator sends SCSI commands REQUEST SENSE, INQUIRY, or GET WINDOW PARAMETERS, it can then ask for either an odd or an even number of bytes of data.

Table 7-2 Transfer data types and transfer identification

Data type code	Data type qualifier	Description
$00	$00	Image data
$01	$02	3-by-3 color correction matrix
$03	$02	Gamma function table
$80	$02	PSRAM table for red/green/blue

The internal hardware structure of the scanner requires that each color of each line, or each line, in the case of monochrome modes, is composed of an even number of bytes. If the scan area requested results in a line that is not an even number of bytes in length, the scanner increases the scan width so that the scan line is an even number of bytes. Figure 7-36 illustrates what happens in this case. The host then adds the additional number of bytes, color, and lines to the originally requested scan width when reading the image data.

Figure 7-36 Extra bit or byte returned for different resolutions and composition modes

Number of pixels per line	Image composition monochrome	Additional bit or byte per line	Image composition color	Additional bit or byte			
				R	G	B	Total
16N	1 bit	0 bits	3 bit	0 bits	0 bits	0 bits	0 bits
16N+1		15 bits (15 pixels)		15 bits	15 bits	15 bits	45 bytes (15 pixels)
16N+2		14 bits (14 pixels)		14 bits	14 bits	14 bits	42 bits (14 pixels)
16N+3		13 bits (13 pixels)		13 bits	13 bits	13 bits	39 bits (13 pixels)
16N+4		12 bits (12 pixels)		12 bits	12 bits	12 bits	36 bits (12 pixels)
16N+5		11 bits (11 pixels)		11 bits	11 bits	11 bits	33 bits (11 pixels)
16N+6		10 bits (10 pixels)		10 bits	10 bits	10 bits	30 bits (10 pixels)
16N+7		9 bits (9 pixels)		9 bits	9 bits	9 bits	27 bits (9 pixels)
16N+8		8 bits (8 pixels)		8 bits	8 bits	8 bits	24 bits (8 pixels)
16N+9		7 bits (7 pixels)		7 bits	7 bits	7 bits	21 bits (7 pixels)
16N+10		6 bits (6 pixels)		6 bits	6 bits	6 bits	18 bits (6 pixels)
16N+11		5 bits (5 pixels)		5 bits	5 bits	5 bits	15 bits (5 pixels)
16N+12		4 bits (4 pixels)		4 bits	4 bits	4 bits	12 bits (4 pixels)
16N+13		3 bits (3 pixels)		3 bits	3 bits	3 bits	9 bits (3 pixels)
16N+14		2 bits (2 pixels)		2 bits	2 bits	2 bits	6 bits (2 pixels)
16N+15		1 bits (1 pixels)		1 bits	1 bits	1 bits	3 bits (1 pixel)
4N	4 bits	0 bits					
4N+1		12 bits (3 pixels)					
4N+2		8 bits (2 pixels)					
4N+3		4 bits (1 pixels)					
8N	8 bits	0 bytes	24 bits	0 bytes	0 bytes	0 bytes	0 bytes
8B+1				1 byte	1 byte	1 byte	3 bytes (1 pixel)

Structure details

Data type codes

Code	Description
00h	Image
01h	Vendor-specific
02h	Halftone mask
03h	Gamma function
04h - 7Fh	Reserved
80h - FFh	Vendor-specific

SEND ($2A)

The SEND command provides a means for the scanner to receive parameter data from the host computer. With the Apple Scanner, you can use this command to download halftone matrixes. With the OneScanner, you can use this command to download halftone matrixes or to set the contents of the scanner's calibration RAM. The Color OneScanner downloads a 3-by-3 matrix for color correction, the gamma table, and PSRAM data write. This section describes the SEND command structures for the Apple Scanner, the Apple OneScanner, and the Apple Color OneScanner. It also details the halftone matrix format for the Scanner and the OneScanner, and the Color OneScanner's 3-by-3 matrix, gamma table, and PSRAM data write.

SEND Command Structure for the Apple Scanner

Figure 7-37 shows the format of the SEND command structure for the Apple Scanner.
Figure 7-37 The SEND command structure for the Apple Scanner

Bit / Byte	7	6	5	4	3	2	1	0
0	($2A) Operation code							
1	Logical unit number			($00) Reserved				
2	($02) Data type code							
3	($00) Reserved							
4	($00) Data type qualifier (MSB)							
5	($02) Data type qualifier (LSB)							
6	Transfer length (MSB)							
7	Transfer length							
8	($11) Transfer length (LSB)							
9	Control							

FIELD DESCRIPTIONS
Here is a list of the fields used in this command structure:

Operation code
The operation code for the SEND command is $2A.

Reserved
These fields are reserved for future expansion. Set them to 0.

Data type code
This field indicates the type of parameter data to be transferred to the scanner. Only halftone parameter data may be transferred to the Apple Scanner. Set this field to $02.

Data type qualifer
This field indicates which halftone matrix within the scanner will be altered. This field must always be $02.

Transfer length
This field indicates the amount of parameter data to be sent. Since the Apple Scanner supports this command only for 4-by-4 halftone matrixes, this parameter must always be set to 17 ($11) — the size of a 4-by-4 halftone parameter page.

SEND Command Structure for the OneScanner

Figure 7-38 shows the format of the SEND command structure for the OneScanner.

Figure 7-38 The SEND command structure for the OneScanner

Bit Byte	7	6	5	4	3	2	1	0
0	($2A) Operation code							
1	Logical unit number			($00) Reserved				
2	($02 or $80) Data type code							
3	($00) Reserved							
4	Data type qualifier (MSB)							
5	Data type qualifier (LSB)							
6	Transfer length (MSB)							
7	Transfer length							
8	Transfer length (LSB)							
9	Control							

FIELD DESCRIPTIONS

Here is a list of the fields used in this command structure:

Operation code
The operation code for the SEND command is $2A.

Reserved These fields are reserved for future expansion. Set them to 0.

Data type code
This field indicates the type of parameter data to be transferred to the scanner. Set this field to one of the following values:
$02 Send halftone pattern to static RAM
$80 Set calibration RAM

Data type qualifer
This field indicates which halftone matrix within the scanner will be altered. This field must always be $02.

Transfer length
This field indicates the amount of parameter data to be sent. For halftone matrixes, set this field to the size of the matrix plus the dimension byte. (See "Halftone Parameter Page Description," later in this chapter, for a detailed description of the format of a halftone matrix.) The OneScanner supports both 4-by-4 and 8-by-8 halftone matrixes.
When setting the contents of the calibration RAM, you must set Transfer length to 2550. The parameter data consists of a contiguous buffer of 2550 bytes. Each byte of parameter data contains the new calibration data for the corresponding pixel sensor in the CCD array.

SEND Command Structure for the Color OneScanner

Figure 7-39 shows the format of the SEND command structure for the Color OneScanner.
Figure 7-39 The SEND command structure for the Color OneScanner

Bit Byte	7	6	5	4	3	2	1	0
0	($2A) Operation code							
1	Logical unit number			($00) Reserved				
2	($01 or $03 or $80) Data type code							
3	($00) Reserved							
4	($00) Data type qualifier (MSB)							
5	($02) Data type qualifier (LSB)							
6	Transfer length (MSB)							
7	Transfer length							
8	Transfer length (LSB)							
9	Control							

FIELD DESCRIPTIONS
Here is a list of the fields used in this command structure:

Operation code
The operation code for the SEND command is $2A.

Reserved
These fields are reserved for future expansion. Set them to 0.

Data type code
This field indicates the type of parameter data to be transferred to the scanner. Set this field to one of the following values:
$01 Send 3-by-3 matrix for color correction
$03 Send gamma table
$80 Set calibration RAM

Data type qualifer
This field is always set to $02.

Transfer length
This field indicates the amount of parameter data to be sent.
If the transfer type is $80 (set calibration RAM), you must set the transfer length to 8100 bytes, to update the PSRAM calibration data for all three lines of sensors. Each single byte of data contains the new calibration data (CD) for each CD pixel sensor in ascending order. Set the transfer length to 18 for the 3-by-3 matrix, and to 768 for the gamma correction table.

Halftone Parameter Page Description for the Apple Scanner and the OneScanner

The halftone parameter page, shown in Figure 7-40, defines the halftone matrix that the application program downloads to the scanner. The size of the parameter page varies according to the dimensions of the halftone matrix. The parameter page contains a single byte that indicates the matrix dimensions, followed by 1 byte of data for each matrix element. Thus, for a 4-by-4 matrix, the parameter page contains 17 bytes of data (the dimension byte followed by 16 bytes of matrix data), the parameter page for an 8-by-8 matrix contains 65 bytes, and so on.

Figure 7-40 The SEND command halftone parameter page

Bit / Byte	7	6	5	4	3	2	1	0
0	Matrix vertical size				Matrix horizontal size			
1	Pixel 0							
2	Pixel 1							
3	Pixel 2							
n + 1	Pixel n							

Matrix vertical size, Matrix horizontal size

These fields indicate the dimensions of the matrix. The dimensions are encoded in a single byte, as follows: bits 4 through 7 contain the y dimension and bits 0 through 3 contain the x dimension. For a 4-by-4 matrix, set this field to $44. For an 8-by-8 matrix, set this field to $88. Figure 7-41 shows how the pixel numbers map onto a 4-by-4 matrix; pixels map similarly for an 8-by-8 matrix, except that there are eight rows, and each row contains eight elements.

Pixels 0 through n

Each of these pixel fields specifies a threshold level at which a particular pixel changes from black to white. Figure 7-41 shows the position of each pixel in a 4-by-4 matrix. The range of threshold values is from 0 to 255. In this range, 255 is the highest threshold setting, 0 is the lowest threshold setting, and 128 is the average threshold setting. A high threshold value results in most gray shades being changed to white. A low threshold value results in most gray shades being changed to black.

Figure 7-41 The halftone matrix pattern for a 4-by-4 matrix

0	1	2	3
4	5	6	7
8	9	10	11
12	13	14	15

Halftone matrix patterns

The following tables contain Spiral and Bayer ordered dither matrix patterns of different sizes.

Spiral 2-by-2 matrix

0	1
3	2

Bayer ordered dither 2-by-2 matrix

0	3
2	1

Spiral 3-by-3 matrix

0	1	2
7	8	3
6	5	4

There is no Bayer ordered dither 3-by-3 matrix as Bayer matrices are square with a side length that is a power of 2.

Spiral 4-by-4 matrix

0	1	2	3
11	12	13	4
10	15	14	5
9	8	7	6

Bayer ordered dither 4-by-4 matrix

0	12	3	15
8	4	11	7
2	14	1	13
10	6	9	5

Spiral 8-by-8 matrix

0	1	2	3	4	5	6	7
27	28	29	30	31	32	33	8
26	47	48	49	50	51	34	9
25	46	59	60	61	52	35	10
24	45	58	63	62	53	36	11
23	44	57	56	55	54	37	12
22	43	42	41	40	39	38	13
21	20	19	18	17	16	15	14

Bayer ordered dither 8-by-8 matrix

0	48	12	60	3	51	15	63
32	16	44	28	35	19	47	31
8	56	4	52	11	59	7	55
40	24	36	20	43	27	39	23
2	50	14	62	1	49	13	61
34	18	46	30	33	17	45	29
10	58	6	54	9	57	5	53
42	26	38	22	41	25	37	21

Gamma Data Write Format for the Color OneScanner

Figure 7-42 shows the gamma data write format.

Figure 7-42 Gamma data write format

Bit Byte	7	6	5	4	3	2	1	0
0	Red gamma value for grayscale value of 0							
1	Red gamma value for grayscale value of 1							
2	Red gamma value for grayscale value of 2							
255	Red gamma value for grayscale value of 255							
256	Green gamma value for grayscale value of 0							
257	Green gamma value for grayscale value of 1							
258	Green gamma value for grayscale value of 2							
511	Green gamma value for grayscale value of 255							
512	Blue gamma value for grayscale value of 0							
513	Blue gamma value for grayscale value of 1							
514	Blue gamma value for grayscale value of 2							
767	Blue gamma value for grayscale value of 255							

Color Correction Matrix for the Color OneScanner

Figure 7-43 shows the color correction matrix.

Figure 7-43 Color correction matrix

Bit Byte	7	6	5	4	3	2	1	0
0	K0 byte 1 (Ka)							
1	K0 byte 2 (Kb)							
2	K1 byte 1 (Ka)							
3	K1 byte 2 (Kb)							
4	K2 byte 1 (Ka)							
5	K2 byte 2 (Kb)							
6	K3 byte 1 (Ka)							
7	K3 byte 2 (Kb)							
8	K4 byte 1 (Ka)							
9	K4 byte 2 (Kb)							
10	K5 byte 1 (Ka)							
11	K5 byte 2 (Kb)							
12	K6 byte 1 (Ka)							
13	K6 byte 2 (Kb)							
14	K7 byte 1 (Ka)							
15	K7 byte 2 (Kb)							
16	K8 byte 1 (Ka)							
17	K8 byte 2 (Kb)							

The format for each byte is shown below. Ka corresponds to byte 1 of K0-8, and Kb corresponds to byte 2 of K0-8. The sign bit indicates plus or minus, MSB is the most significant bit, LSB the least significant bit.

	Ka		Kb		
Byte	Bit 15	Bits 9 - 14	Bit 8	Bits 7 - 1	Bit 0
Ka and Kb	Sign	0	MSB	Data bits	LSB

PSRAM Data Write Format for the Color OneScanner

Figure 7-44 shows the PSRAM data write format.

Figure 7-44 PSRAM data write format

Bit Byte	7	6	5	4	3	2	1	0
0	Calibration data for red CCD pixel sensor #1							
1	Calibration data for red CCD pixel sensor #2							
2	Calibration data for red CCD pixel sensor #3							
2699	Calibration data for red CCD pixel sensor #2700							
2700	Calibration data for green CCD pixel sensor #1							
2701	Calibration data for green CCD pixel sensor #2							
2702	Calibration data for green CCD pixel sensor #3							
5399	Calibration data for green CCD pixel sensor #2700							
5400	Calibration data for blue CCD pixel sensor #1							
5401	Calibration data for blue CCD pixel sensor #2							
5402	Calibration data for blue CCD pixel sensor #3							
8099	Calibration data for blue CCD pixel sensor #2700							

OBJECT POSITION ($31)

The OBJECT POSITION command allows the host computer to control the position of the scanner carriage assembly. The command provides options to park and unpark the carriage and to set the carriage to a specified position. Figure 7-45 shows the format of the OBJECT POSITION command structure.

Note
This command is valid only for the OneScanner and the Color OneScanner. The Apple Scanner does not support this command.

Figure 7-45 The OBJECT POSITION command structure

Bit Byte	7	6	5	4	3	2	1	0
0	($31) Operation code							
1	Logical unit number			($00) Reserved		Position function		
2	Count (MSB)							
3	Count							
4	Count (LSB)							
5 - 8	($00) Reserved							
9	Control							

FIELD DESCRIPTIONS
Here is a list of the fields used in this command structure:

Operation code
: The operation code for the OBJECT POSITION command is $31.

Reserved
: These fields are reserved for future expansion. Set them to 0.

Position type
: This field specifies the action for the scanner to take. Valid values are as follows:

  Unpark carriage $00
  : Instructs the scanner to move the carriage to the home position. If the carriage is already at the home position, the scanner returns a GOOD status. If the scanner cannot move the carriage to the home position, the scanner returns a CHECK CONDITION status and sets the sense data to VENDOR UNIQUE ($9).

  Park carriage $01
  : Instructs the scanner to move the carriage to the shipment lock position. If the carriage is already at that position, the scanner returns a GOOD status.

  Absolute position $02
  : Instructs the scanner to position the carriage at the scan line specified in the Count field. A 0 value causes the scanner to place the carriage at the beginning of the scan area. Other values must indicate a valid y-axis position in increments of 1/1200 of an inch. If the value of Count corresponds to an invalid scan line, the scanner returns a CHECK CONDITION status and sets the sense data to ILLEGAL REQUEST ($5).

  Relative position $03
  : Instructs the scanner to move the carriage to a position relative to its current position. The value of the Count field indicates the direction and distance for the operation. Positive values move the carriage forward; negative values move the carriage backward. The distance is expressed in increments of 1/1200 of an inch. A 0 value does not move the carriage. If the value of Count yields an invalid scan line, the scanner returns a CHECK CONDITION status and sets the sense data to ILLEGAL REQUEST ($5).

  Rotate object $04
  : This position function specifies that the object is rotated in an anti-clockwise direction expressed in thousandths of a degree. The count field specifies the number of units that the object is to be moved.

  $05 - $07
  : The three remaining codes are reserved.

Count
: This field specifies the distance for absolute and relative position requests. The scanner ignores this field for park and unpark requests.

GET DATA BUFFER STATUS ($34)

The GET DATA BUFFER STATUS command allows the host computer to determine how much image data the scanner currently holds. The host computer can then decide whether to issue a READ command to retrieve the stored data. The scanner reports data availability for those windows that were defined with the DEFINE WINDOW PARAMETERS command and were specified in the current SCAN command. Figure 7-46 shows the format of the GET DATA STATUS command structure. The returned data blocks for each supported scanner are described later in this section.

Figure 7-46 The GET DATA STATUS command structure

Bit Byte	7	6	5	4	3	2	1	0
0	($34) Operation code							
1	Logical unit number			($00) Reserved				Wait
2 - 5	($00) Reserved							
6	Allocation length (MSB)							
7	Allocation length							
8	Allocation length (LSB)							
9	Control							

FIELD DESCRIPTIONS
Here is a list of the fields and the bits used in this command structure:

Operation code
: The operation code for the GET DATA BUFFER STATUS command is $34.

Reserved
: These fields are reserved for future expansion. Set them to 0.

Wait
: This bit indicates when the scanner returns data status to the host computer. If the value of the field is 1, the scanner waits for the quantity of data in the scanner's internal memory to reach the limit set by the MODE SELECT command before the scanner responds with data. (This limit is set by the Buffer full ratio field of the disconnect-reconnect parameter page.) If the value is 0, the scanner responds immediately.

Allocation length
: This field specifies the number of bytes that the host computer has allocated for returned data status. If the value of the Allocation length is 0, the scanner will not return data status (this is not an error condition). With any other value, the maximum number of bytes is transferred. The command terminates when the bytes specified by the Allocation length field have been transferred, or when all available status data bytes have been transferred to the host, whichever comes first.

GET DATA BUFFER STATUS Return Structure Description for the Apple Scanner

Figure 7-47 shows the format of the GET DATA BUFFER STATUS return structure for the Apple Scanner.
Figure 7-47 The GET DATA BUFFER STATUS return structure for the Apple Scanner

Bit Byte	7	6	5	4	3	2	1	0
0	($00) Data buffer status length (MSB)							
1	($00) Data buffer status length							
2	($08) Data buffer status length (LSB)							
3	($00) Reserved							Block

Bit Byte	7	6	5	4	3	2	1	0
0	Window identifier							
1	($00) Reserved							
2	Available data buffer (MSB)							
3	Available data buffer							
4	Available data buffer (LSB)							
5	Filled data buffer (MSB)							
6	Filled data buffer							
7	Filled data buffer (LSB)							

FIELD DESCRIPTIONS

The parameter list header describes the length, in bytes, of the descriptor information that follows. Here are the parameter list fields used in this structure:

Data length — This field indicates the number of bytes of status data that follows, starting at byte 4. The value of this field does not include bytes 0 through 3. Each return structure is associated with a window descriptor. If the Apple bit is set in the DEFINE WINDOW PARAMETERS command, the scanner returns data status for only the first window descriptor. Data for secondary scan areas is returned together with the data for the primary scan area. The scanner indicates that a scan is complete and that all data has been transferred by setting the Data length field to 0 and sending no further status data.

Reserved — These fields are reserved for future expansion. Set them to 0.

Block — This field indicates the scanner buffer status. A value of 1 indicates that the scanner's internal buffers are full and that the scanner must transfer all available scan data to the host computer before it can generate more scan data. The bit is also set to 1 when the scanner has reached the end of the scan, even though the buffer may not be full. A value of 0 indicates that the scanner is not currently blocked because of insufficient buffer space.

The data status descriptors contain information about defined scan windows. Each descriptor applies to a single window. Here are the data descriptors used in this structure:

Window identifier
: This field identifies the window associated with this return structure. This value matches the window identifier in the window descriptor parameter of the DEFINE WINDOW PARAMETERS command.

Reserved — These fields are reserved for future expansion. Set them to 0.

Available data buffer
: The available data buffer field indicates, in bytes, the amount of buffer available for transfers from the initiator. This field is valid only in scanners with the ability to accept data from an initiator for processing.

Filled data buffer
: This field indicates the number of bytes of data that the scanner can send for the scan window identified by the Window identifier field. If the value is 0, it indicates that the scanner currently has no data available for the window (perhaps because it has not reached the window in its scanning process). This is not an error condition.

GET DATA BUFFER STATUS Return Structure Description for the OneScanner and the Color OneScanner

Figure 7-48 shows the format of the GET DATA BUFFER STATUS return structure.

Figure 7-48 The GET DATA BUFFER STATUS return structure

Bit Byte	7	6	5	4	3	2	1	0
0	($00) Data buffer status length (MSB)							
1	($00) Data buffer status length							
2	($08) Data buffer status length (LSB)							
3	($00) Reserved							Block

Bit Byte	7	6	5	4	3	2	1	0
0	Window identifier							
1	($00) Reserved							
2	Available data buffer (MSB)							
3	Available data buffer							
4	Available data buffer (LSB)							
5	Filled data buffer (MSB)							
6	Filled data buffer							
7	Filled data buffer (LSB)							

FIELD DESCRIPTIONS

The parameter list header describes the length, in bytes, of the descriptor information that follows. Here are the parameter list fields used in this structure:

Data length
: This field indicates the number of bytes of status data that follows, starting at byte 4. The value of this field does not include bytes 0 through 3. Each return structure is associated with a window descriptor. If the Apple bit is set in the DEFINE WINDOW PARAMETERS command, the scanner returns data status for only the first window descriptor. Data for secondary scan areas is returned together with the data for the primary scan area. The scanner indicates that a scan is complete and that all data has been transferred by setting the Data length field to 0 and sending no further status data.

Reserved
: These fields are reserved for future expansion. Set them to 0.

Block
: This field indicates the scanner buffer status. A value of 1 indicates that the scanner's internal buffers are full and that the scanner must transfer all available scan data to the host computer before it can generate more scan data. The bit is also set to 1 when the scanner has reached the end of the scan, even though the buffer may not be full. A value of 0 indicates that the scanner is not currently blocked because of insufficient buffer space.

The data status descriptors contain information about defined scan windows. Each descriptor applies to a single window. Here are the data descriptors used in this structure:

Window identifier
: This field identifies the window associated with this return structure. This value matches the window identifier in the window descriptor parameter of the DEFINE WINDOW PARAMETERS command.

Reserved
: These fields are reserved for future expansion. Set them to 0.

Available data buffer
: The available data buffer field indicates, in bytes, the amount of buffer available for transfers from the initiator. This field is valid only in scanners with the ability to accept data from an initiator for processing.

Filled data buffer
: This field indicates the number of bytes of data that the scanner can send for the scan window identified by the Window identifier field. If the value is 0, it indicates that the scanner currently has no data available for the window (perhaps because it has not reached the window in its scanning process). This is not an error condition.

UNKNOWN COMMAND $D3 ($D3)

Bit / Byte	7	6	5	4	3	2	1	0
0	($D3) Operation code							
1	Logical unit number			($00) Reserved				
2	($00) Reserved							
3	($00) Reserved							
4	Allocation length							
5	Control							

UNKNOWN COMMAND $D4 ($D4)

Bit Byte	7	6	5	4	3	2	1	0
0	($D4) Operation code							
1	Logical unit number			($00) Reserved				
2	($00) Reserved							
3	($00) Reserved							
4	Allocation length							
5	Control							

UNKNOWN COMMAND $D5 ($D5)

Bit Byte	7	6	5	4	3	2	1	0
0	($D5) Operation code							
1	Logical unit number			($00) Reserved				
2	($00) Reserved							
3	($00) Reserved							
4	Allocation length							
5	Control							

UNKNOWN COMMAND $D6 ($D6)

Bit Byte	7	6	5	4	3	2	1	0
0	($D6) Operation code							
1	Logical unit number			($00) Reserved				
2	($00) Reserved							
3	($00) Reserved							
4	Allocation length							
5	Control							

The programming part

```
Open   =     $2010  ; GS/OS reference
Read   =     $2012
Close  =     $2014
DInfo  =     $202C
DStatus =    $202D
DControl =   $202E

devSCANNER = $001A  ; GS/OS (Device) reference

attrFixed =  $4000  ; Memory Manager
```

Find a scanner device

The routine goes through the list of connected devices and exits with the carry clear when it finds a scanner device. The string for the first device is 'APPLESCSI.SCANNER01.00'

```
       lda    #1            ; start with device 1
       sta    proDINFO+2

]lp    jsl    GSOS          ; do a DInfo
       dw     DInfo
       adrl   proDINFO
       bcc    found         ; we have a device

       cmp    #$0011        ; do we have more devices?
       bne    loop
       rts                  ; no, return with carry set

loop   inc    proDINFO+2    ; check next device
       bra    ]lp

found  lda    proDINFO+20   ; is it a scanner?
       cmp    #devSCANNER
       bne    loop          ; no
       rts                  ; yes, return with carry clear

*---

proDINFO
       dw     8             ; 00 pcount
       ds     2             ; 02 device num
       adrl   devINFO       ; 04 device name
       ds     2             ; 08 characteristics
       ds     4             ; 0A total blocks
       ds     2             ; 0E slot number
       ds     2             ; 10 unit number
       ds     2             ; 12 version
       ds     2             ; 14 device id

devINFO
       dw     $36           ; buffer size (GS/OS output buffer)
devNAME
       ds     $34           ; device name (GS/OS string)
```

Identify the scanner brand and model

Now that we have our device ID, we can use it to call the DControl Inquiry command that will return the information we want.

The scanner brand (Vendor identification) is a left-justified string at inqBUFFER[8..15]. The scanner model (Product identification) is a left-justified string at inqBUFFER[16..31]. See "How to identify" for more information.

```
       lda    proDINFO+2    ; get the device number
       sta    proINQUIRY+2

       jsl    GSOS          ; get the Inquiry table data
```

```
       dw     DStatus
       adrl   proINQUIRY
       rts

*---

proINQUIRY
       dw     5             ; 00 pcount
       ds     2             ; 02 device num
       dw     $8012         ; 04 status code
       adrl   inqLIST       ; 06 status list
       adrl   240           ; 0A request count
       ds     4             ; 0E transfer count

inqLIST
       ds     2             ; always 0000
       hex    12            ; 00 command
       ds     3             ; 01 3 zeroes
       dfb    240           ; 04 length
       ds     7             ; 05 7 zeroes
       adrl   inqBUFFER     ; 12 result buffer

inqBUFFER
       ds     240           ; result buffer, see Table 45 description for more information
```

Preparing the scanner

The first action is to open the device to let the Character FST communicate with the scanner. We also set the scanner in WAIT mode so that the READ call will return to the caller once all data is read.

```
       jsl    GSOS          ; open the device for the CHAR.FST
       dw     Open
       adrl   proOPEN

       lda    proDINFO+2    ; get the device number
       sta    proINQUIRY+2

       jsl    GSOS          ; set the device in WAIT mode
       dw     DControl
       adrl   proWAIT
       rts

*---

proOPEN
       dw     2             ; 00 pcount
       ds     2             ; 02 ref num
       adrl   devNAME       ; 04 pathname

proWAIT
       dw     5             ; 00 pcount
       ds     2             ; 02 device num
       dw     $0004         ; 04 control code
       adrl   waitLIST      ; 06 control list
       adrl   2             ; 0A request count
       ds     4             ; 0E transfer count

waitLIST
       dw     $0000         ; $0000: wait mode / $8000: no-wait mode
```

Scanning a page

Scanning a page means preparing the window that defines the scanning area. Then, we must reserve memory for the resulting picture. Finally, we scan the page.

```
* 1. Prepare the scanning window
```

```
* 2. We set the scanning area

        lda     proDINFO+2    ; get the device number
        sta     proWINDOW+2

        jsl     GSOS          ; define the scanning window
        dw      DControl
        adrl    proWINDOW

* 3. We reserve memory

        pha                   ; space for result
        pha
        PushLong bufferSIZE ; the number of bytes to reserve
        PushWord myID       ; my user ID
        PushWord #attrFixed ; $4000
        PushLong #0         ; location pointer
        _NewHandle
        phd                   ; deference the handle
        tsc                   ; to get the pointer
        tcd
        lda     [3]
        sta     proREAD+4
        ldy     #2
        lda     [3],y
        sta     proREAD+6
        pld
        PullLong myHANDLE

* 4. We scan the page

        lda     proOPEN+2     ; get the ref number
        sta     proREAD+2
        sta     proCLOSE+2

        jsl     GSOS          ; read the data from the scanner
        dw      Read
        adrl    proREAD

        jsl     GSOS          ; close the file
        dw      Close
        adrl    proCLOSE
        rts                   ; we're done!

*---

myID
        ds      2             ; my (your) Memory Manager user ID

bufferSIZE
        ds      4             ; size of data to scan

myHANDLE
        ds      4             ; handle to the scanned area

proWINDOW

proREAD
        dw      4             ; 00 pcount
        ds      2             ; 02 ref num
        ds      4             ; 04 data buffer
        ds      4             ; 08 request count
        ds      4             ; 0A transfer count

proCLOSE
        dw      1             ; 00 pcount
        ds      2             ; 02 ref num
```

References

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See Chapter 15 for scanner devices

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Glossary

Byte	Indicates an 8-bit construct.
Command descriptor block (CDB)	The structure used to communicate commands from an initiator to a target.
Host adapter	A device which connects between a host system and the SCSI bus. The device usually performs the lower layers of the SCSI protocol and normally operates in the initiator role. This function may be integrated into the host system.
Initiator	An SCSI device that requests an I/O process to be performed by another SCSI device (a target).
I_T_L nexus	A nexus which exists between an initiator, a target, and a logical unit. This relationship replaces the prior I_T nexus.
Logical unit	A physical or virtual peripheral device addressable through a target.
Logical unit number	An encoded three-bit identifier for the logical unit.
Nexus	A relationship that begins with the establishment of an initial connection and ends with the completion of the I/O process. The relationship may be restricted to specify a single logical unit or target routine by the successful transfer of an IDENTIFY message. The relationship may be further restricted by the successful transfer of a queue tag message.
Optional	The referenced item is not required to claim compliance with this standard. Implementation of an optional item must be as defined in this standard.
Reserved	Identifies bits, fields, and code values that are set aside for future standardization.
Scanner	Any graphic input device that converts printed matter into bit (binary digital) data.
Status	One byte of information sent from a target to an initiator upon completion of each command.
Target	An SCSI device that performs an operation requested by an initiator.
Vendor-specific	Something (e.g. a bit, field, code value, etc.) that is not defined by this standard and may be used differently in various implementations.